

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

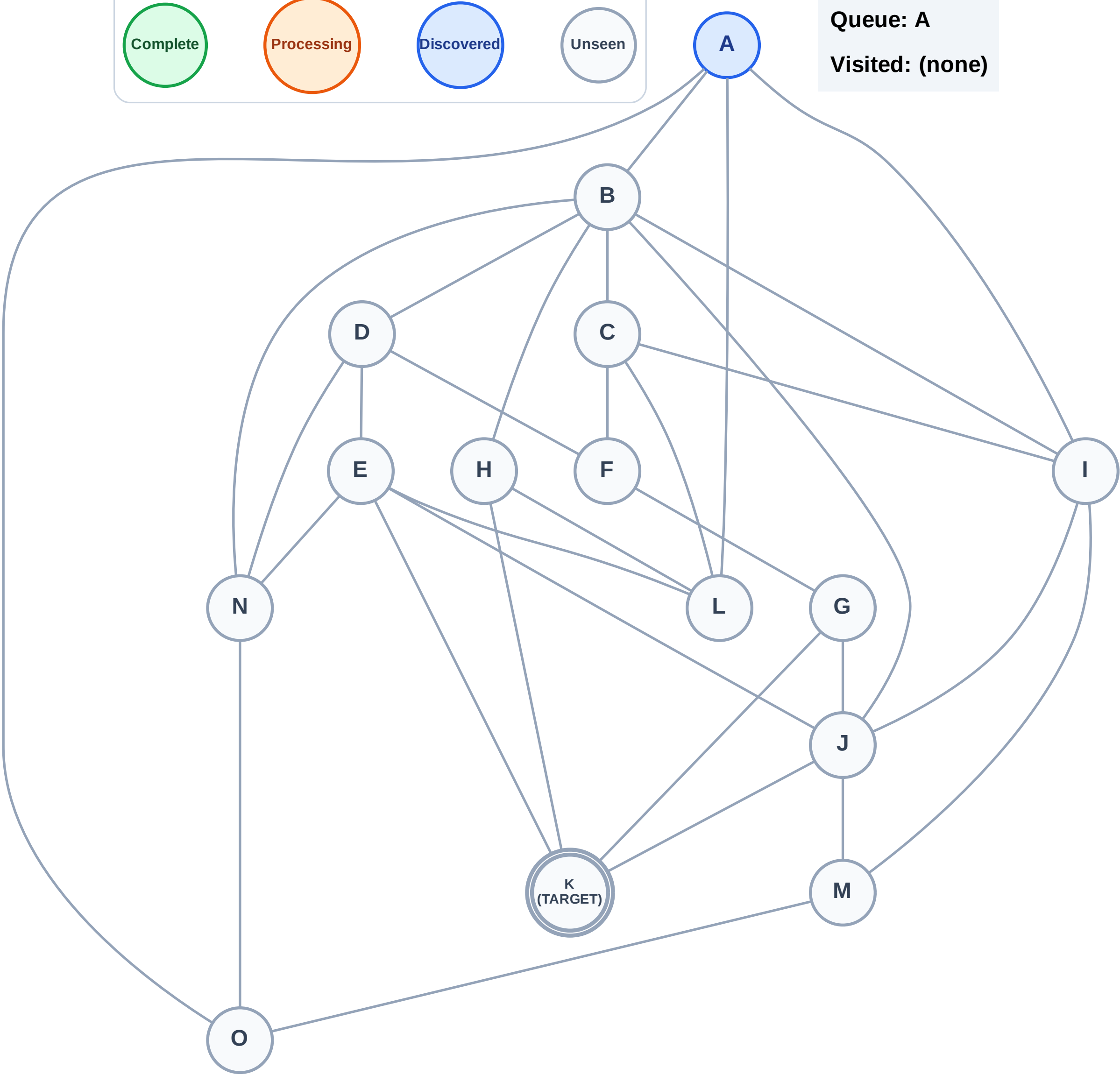
Complete

Processing

Discovered

Unseen

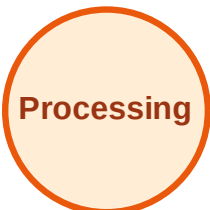
Queue: A
Visited: (none)



BFS Traversal

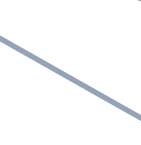
Step 2: Process node A

Legend



Queue: (empty)

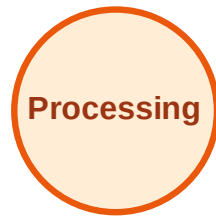
Visited: (none)



BFS Traversal

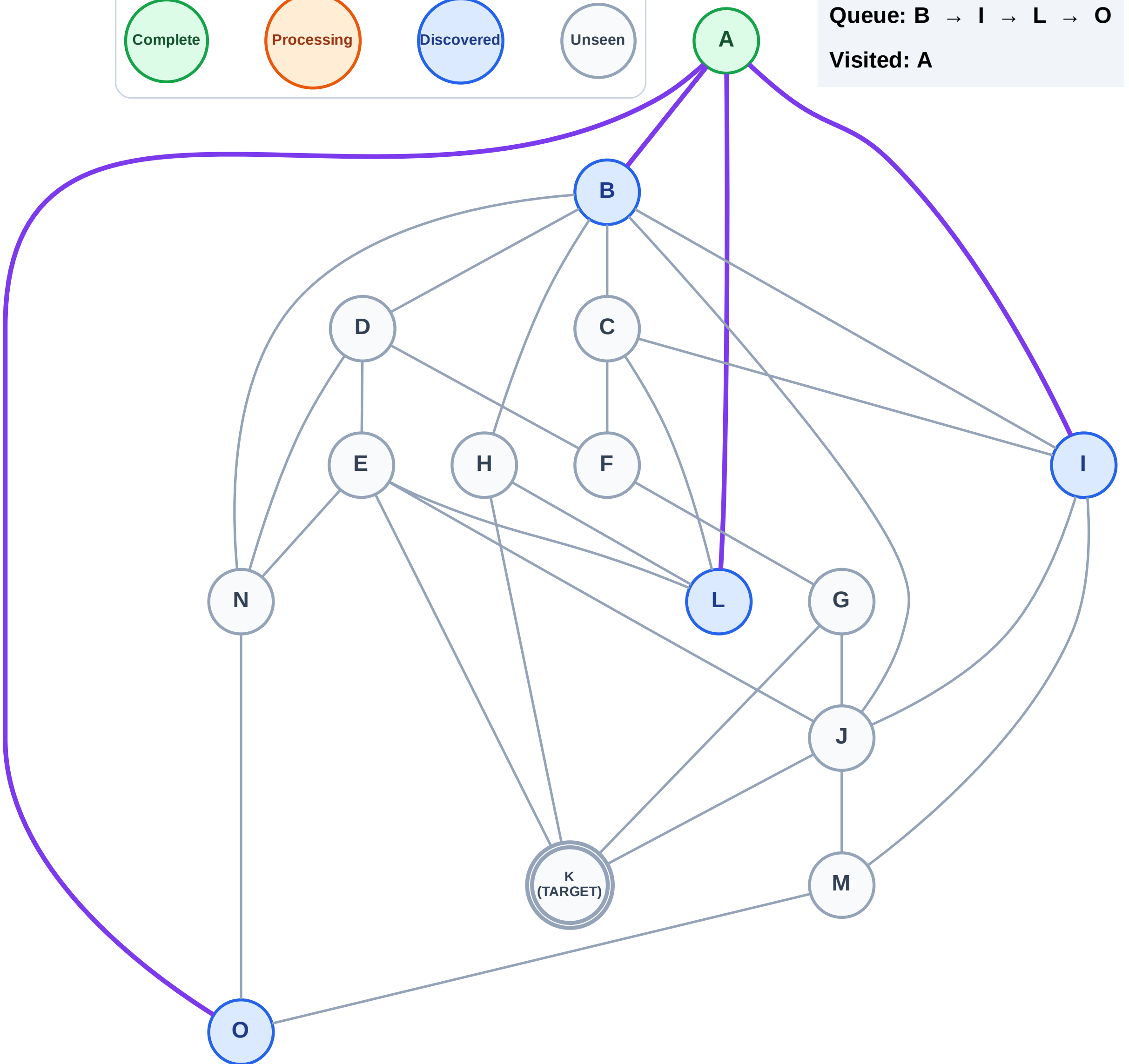
Step 3: Discover B, I, L, O from A

Legend



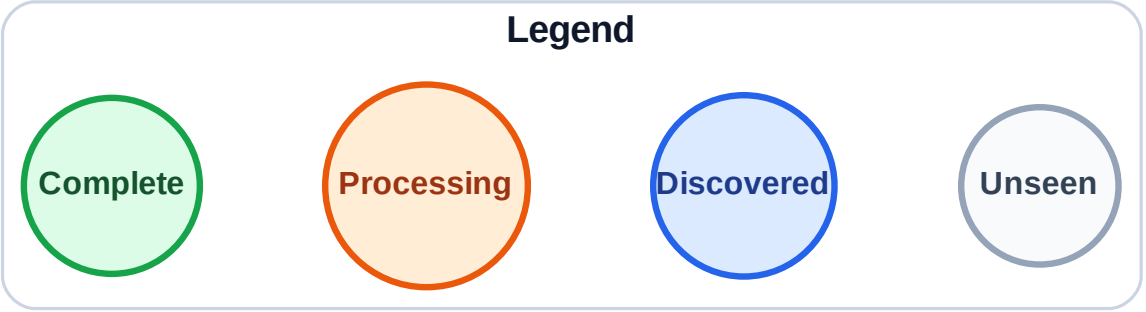
Queue: B → I → L → O

Visited: A

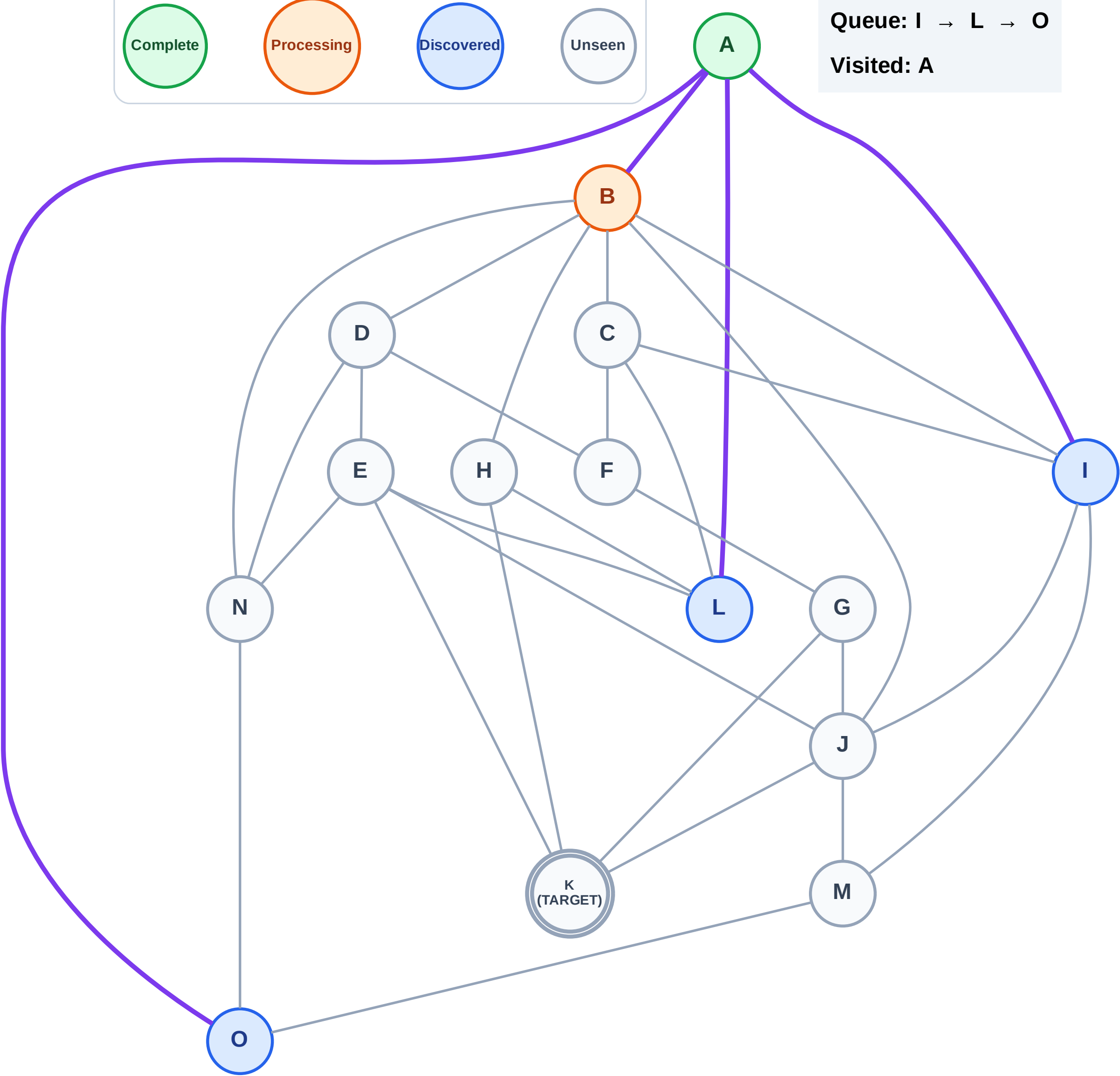


BFS Traversal

Step 4: Process node B



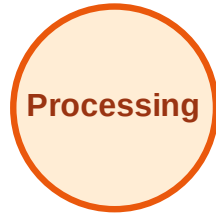
Queue: I → L → O
Visited: A



BFS Traversal

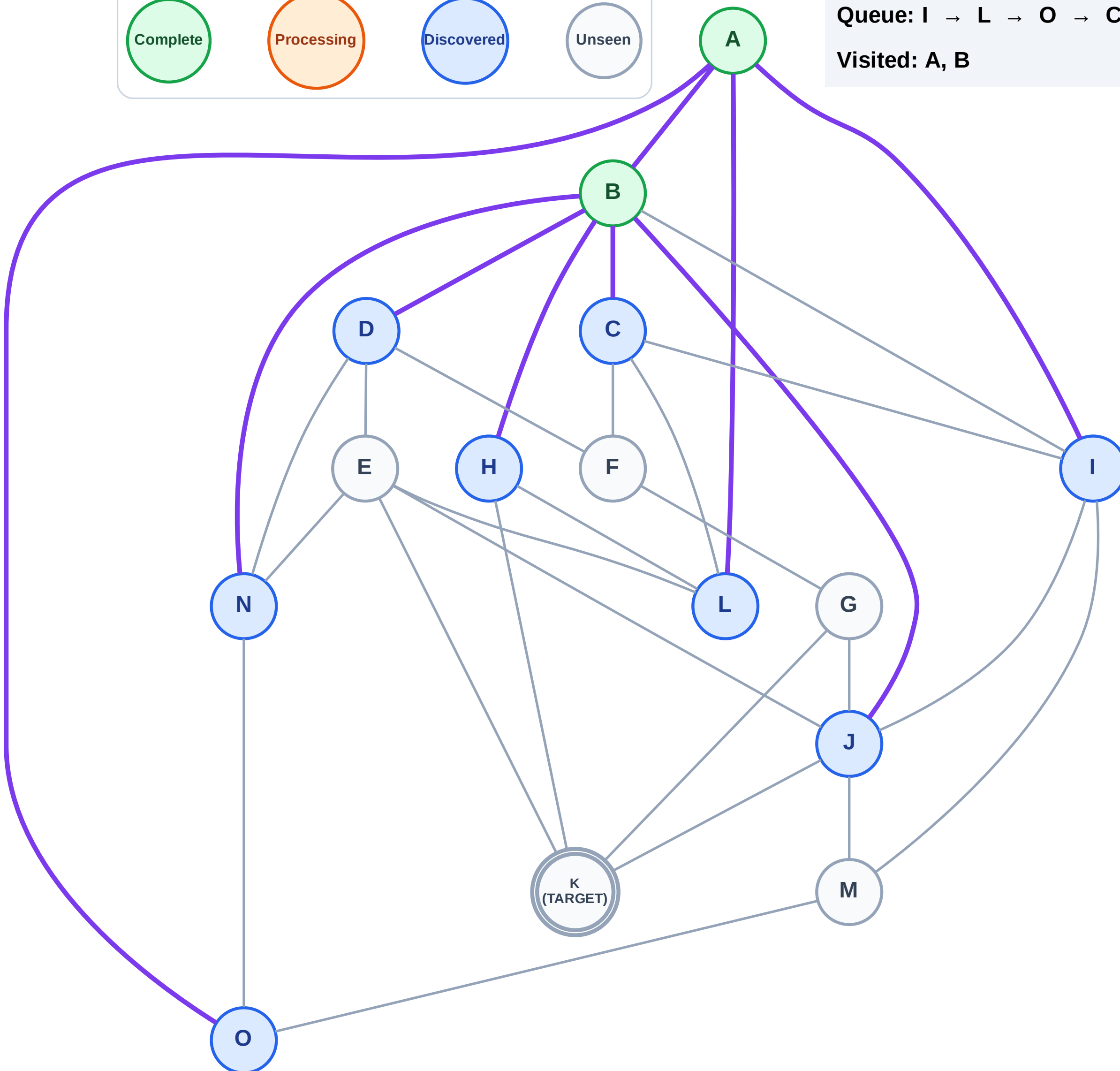
Step 5: Discover C, D, H, J, N from B

Legend



Queue: I → L → O → C → D → H → J → N

Visited: A, B



BFS Traversal

Step 6: Process node I

Legend

Complete

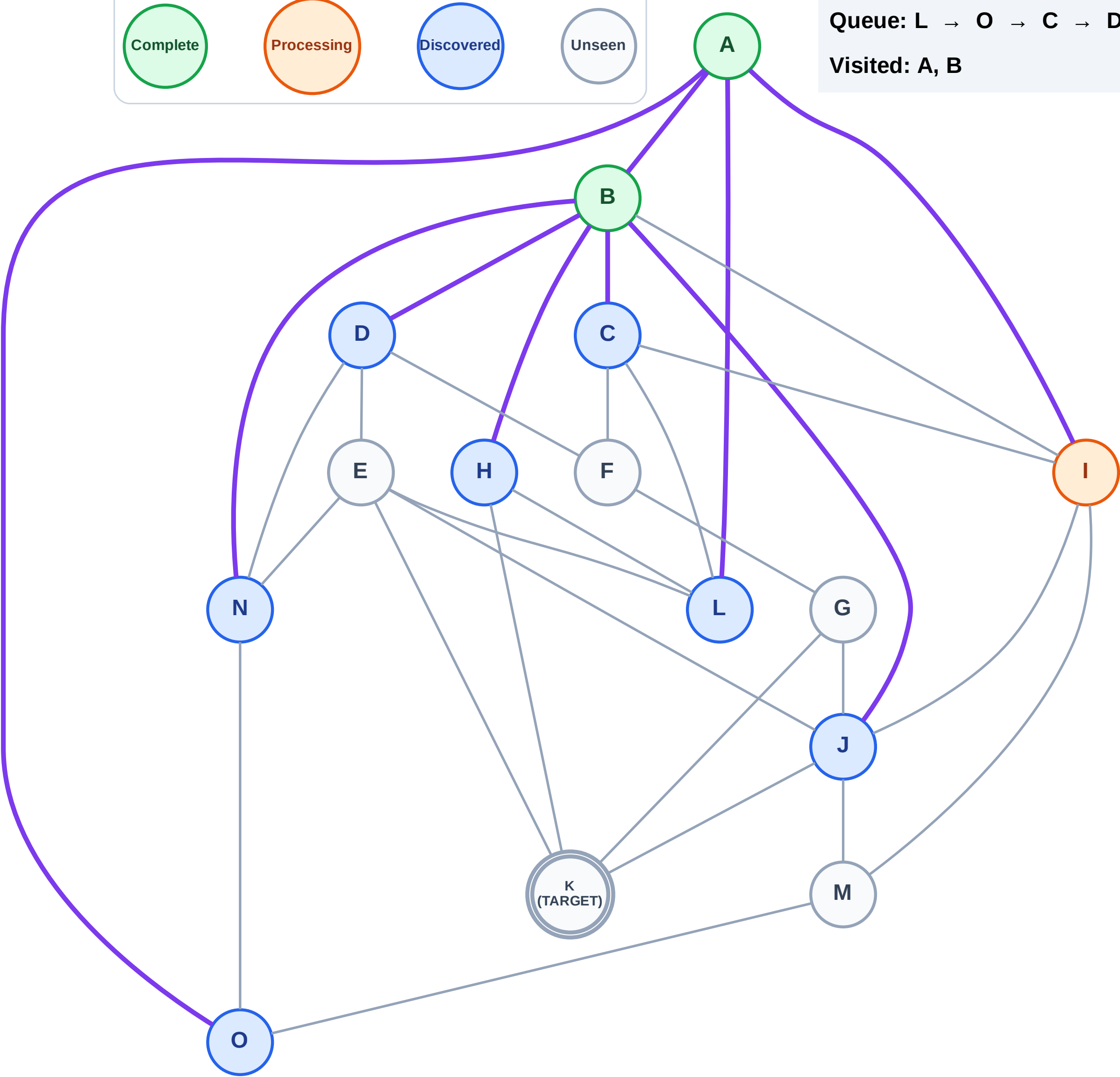
Processing

Discovered

Unseen

Queue: L → O → C → D → H → J → N

Visited: A, B



BFS Traversal

Step 7: Discover M from I

Legend



Complete



Processing



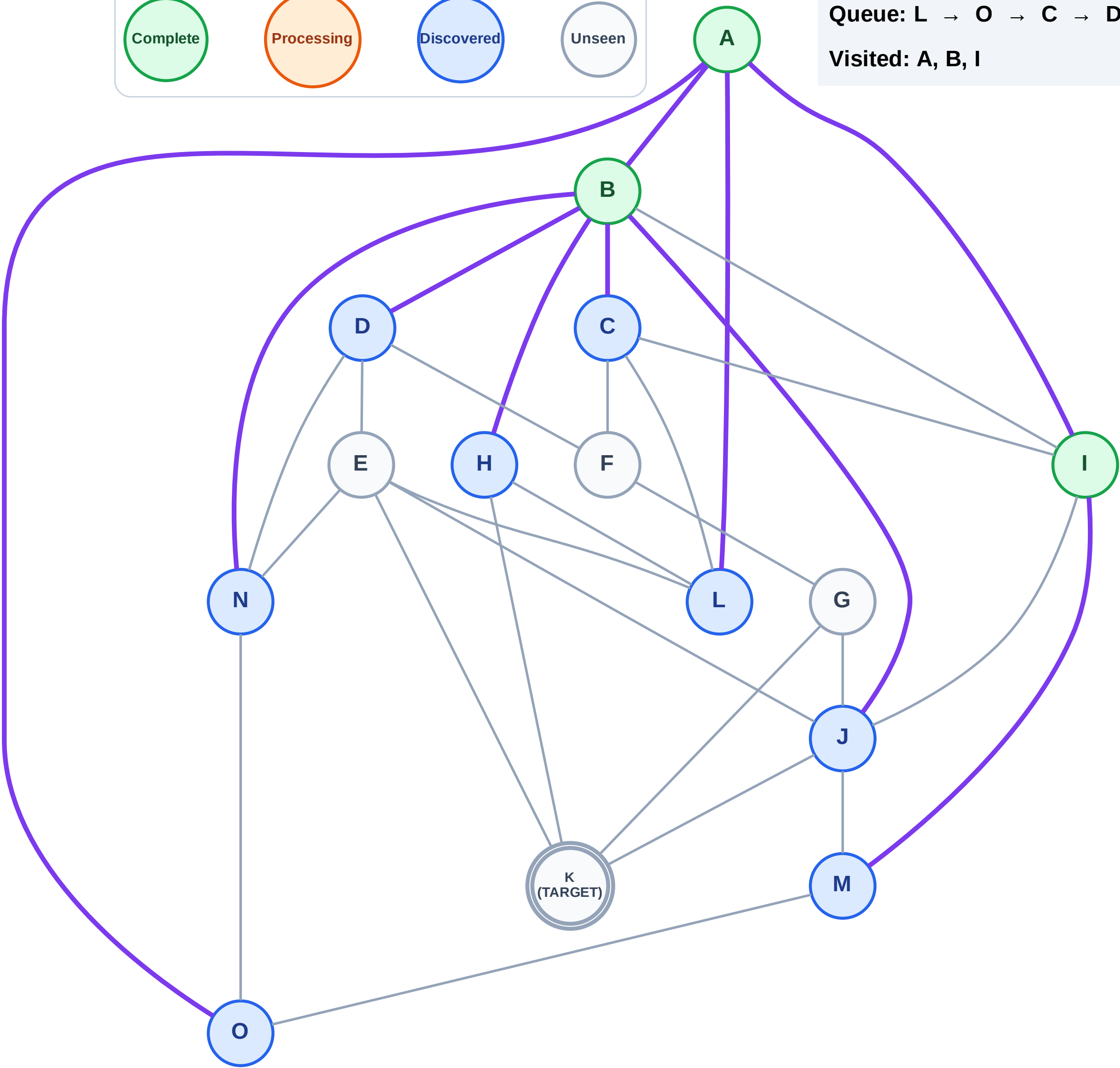
Discovered



Unseen

Queue: L → O → C → D → H → J → N → M

Visited: A, B, I



BFS Traversal

Step 8: Process node L

Legend

Complete

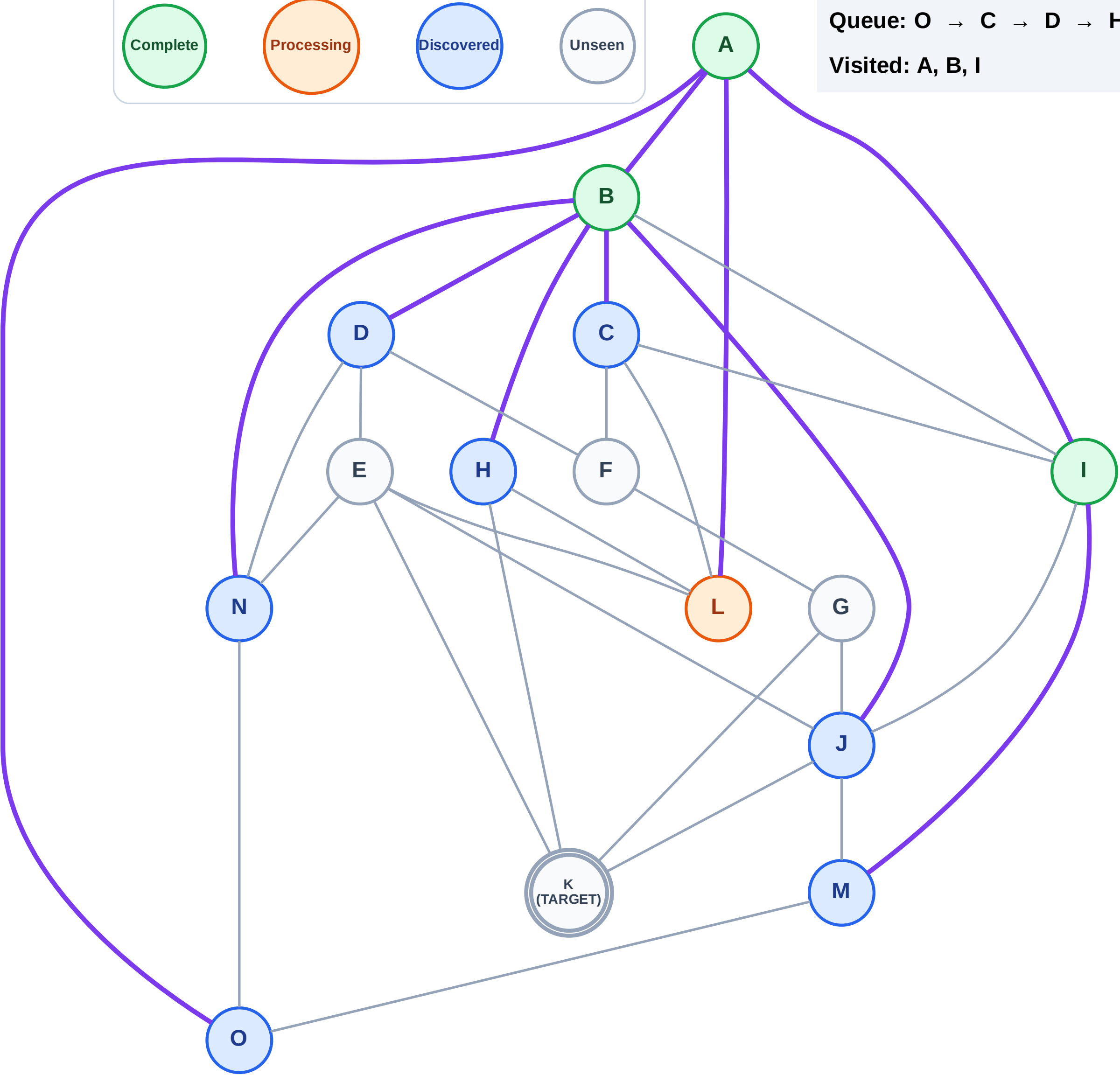
Processing

Discovered

Unseen

Queue: O → C → D → H → J → N → M

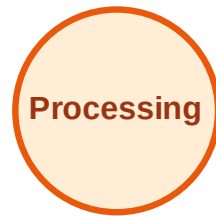
Visited: A, B, I



BFS Traversal

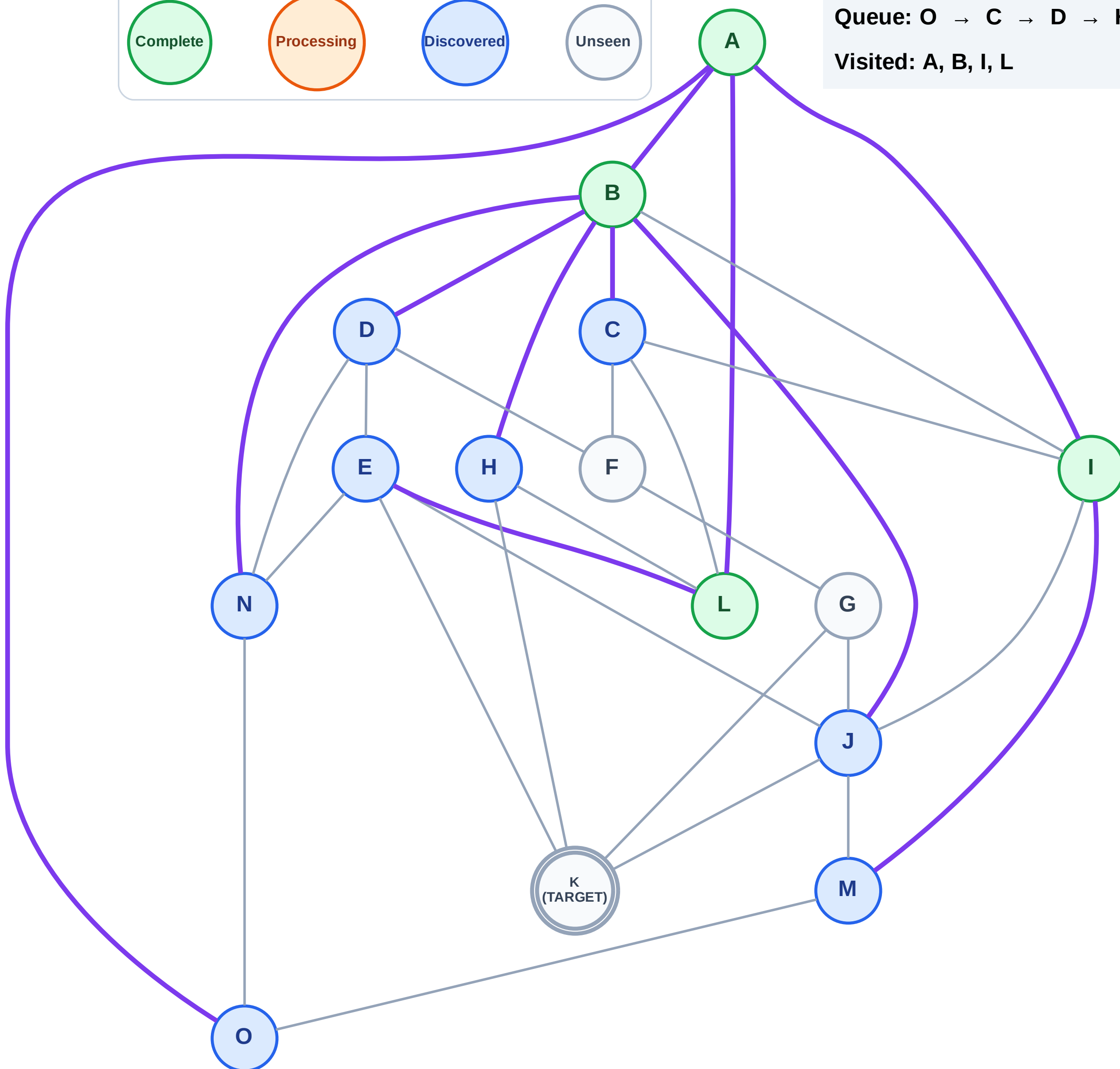
Step 9: Discover E from L

Legend



Queue: O → C → D → H → J → N → M → E

Visited: A, B, I, L



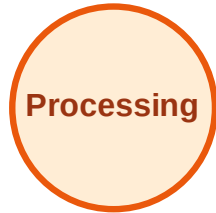
BFS Traversal

Step 10: Process node O

Legend



Complete



Processing



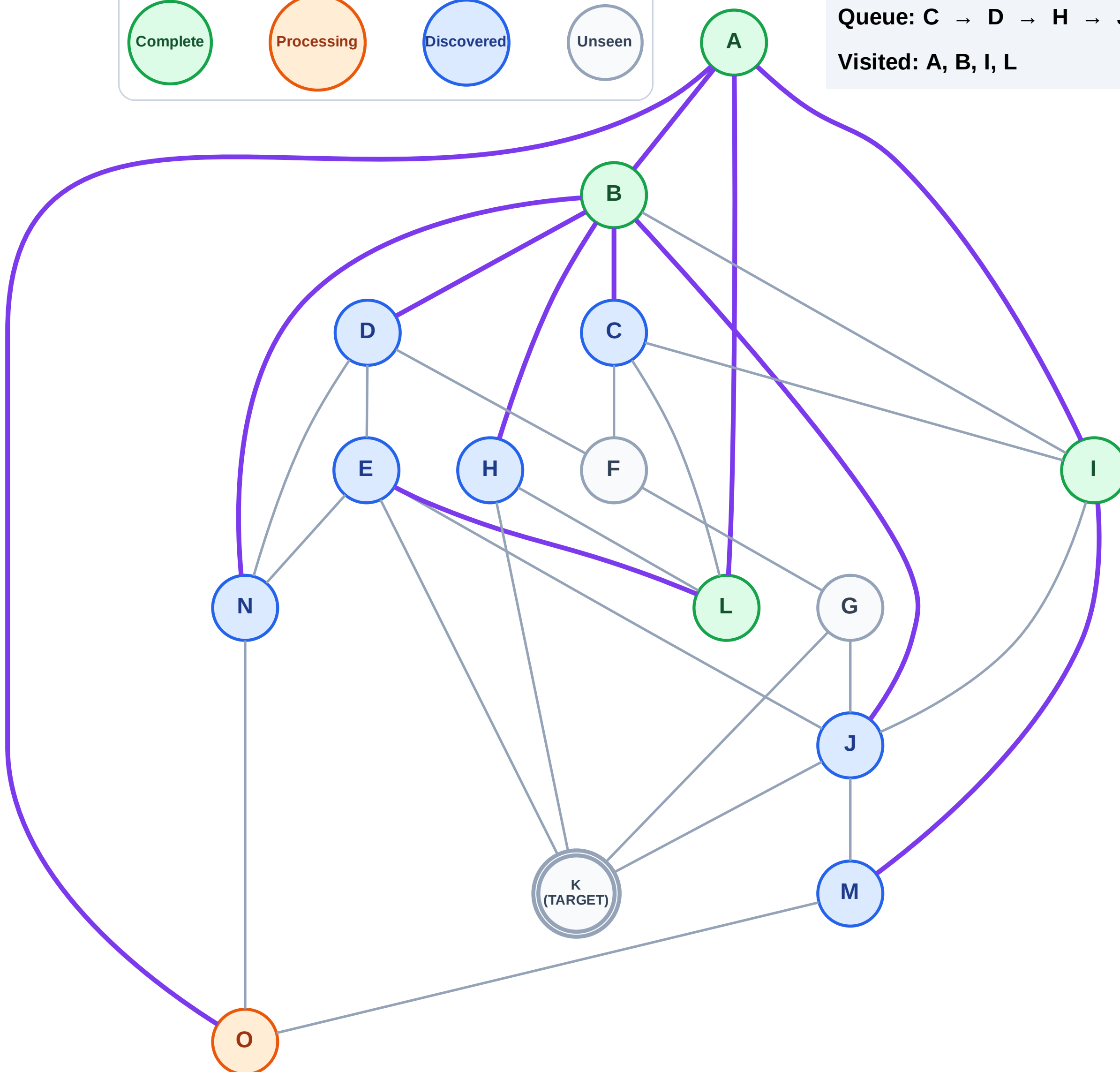
Discovered



Unseen

Queue: C → D → H → J → N → M → E

Visited: A, B, I, L



BFS Traversal

Step 11: No new neighbors from O

Legend

Complete

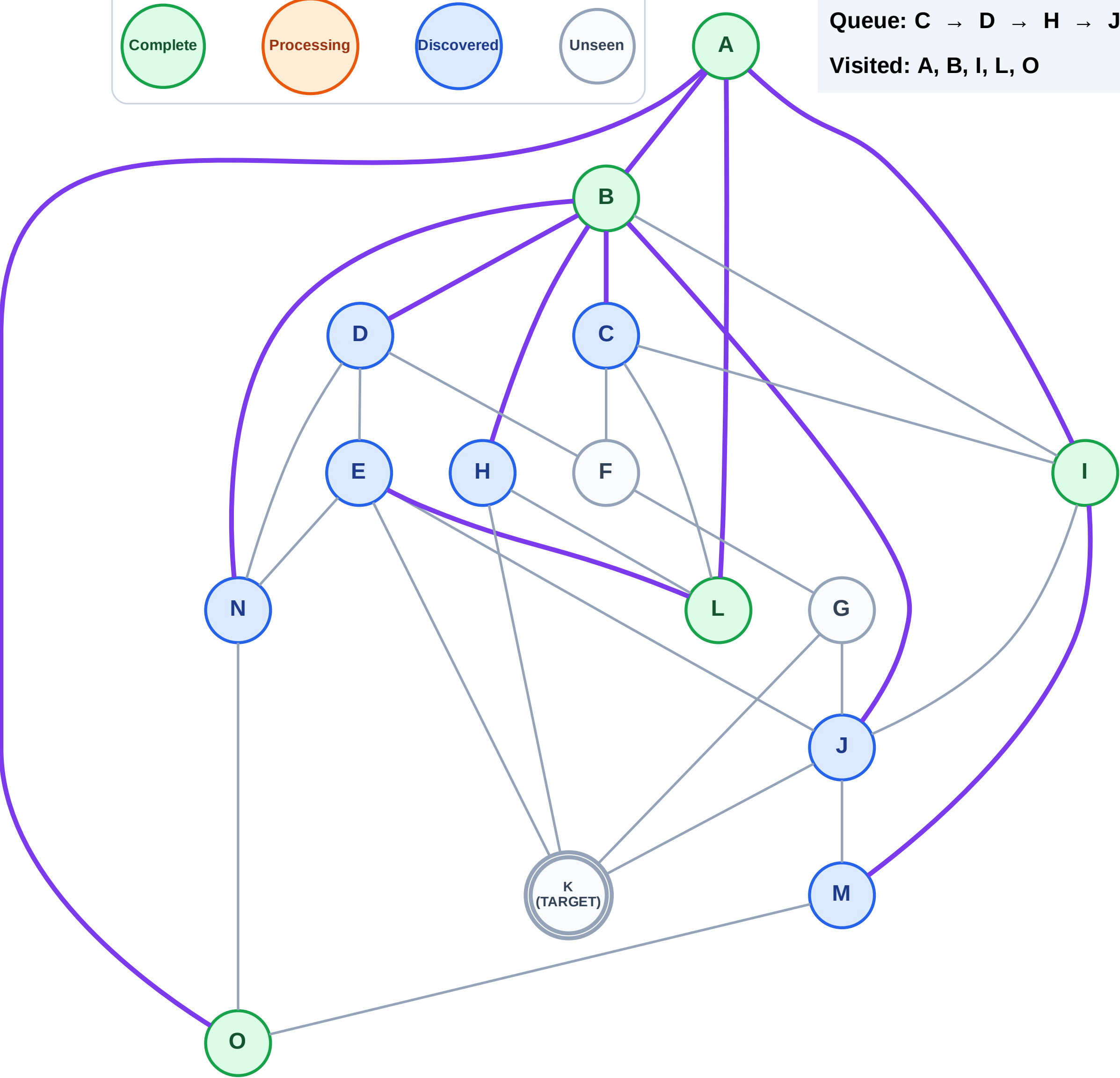
Processing

Discovered

Unseen

Queue: C → D → H → J → N → M → E

Visited: A, B, I, L, O



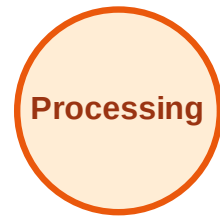
BFS Traversal

Step 12: Process node C

Legend



Complete



Processing



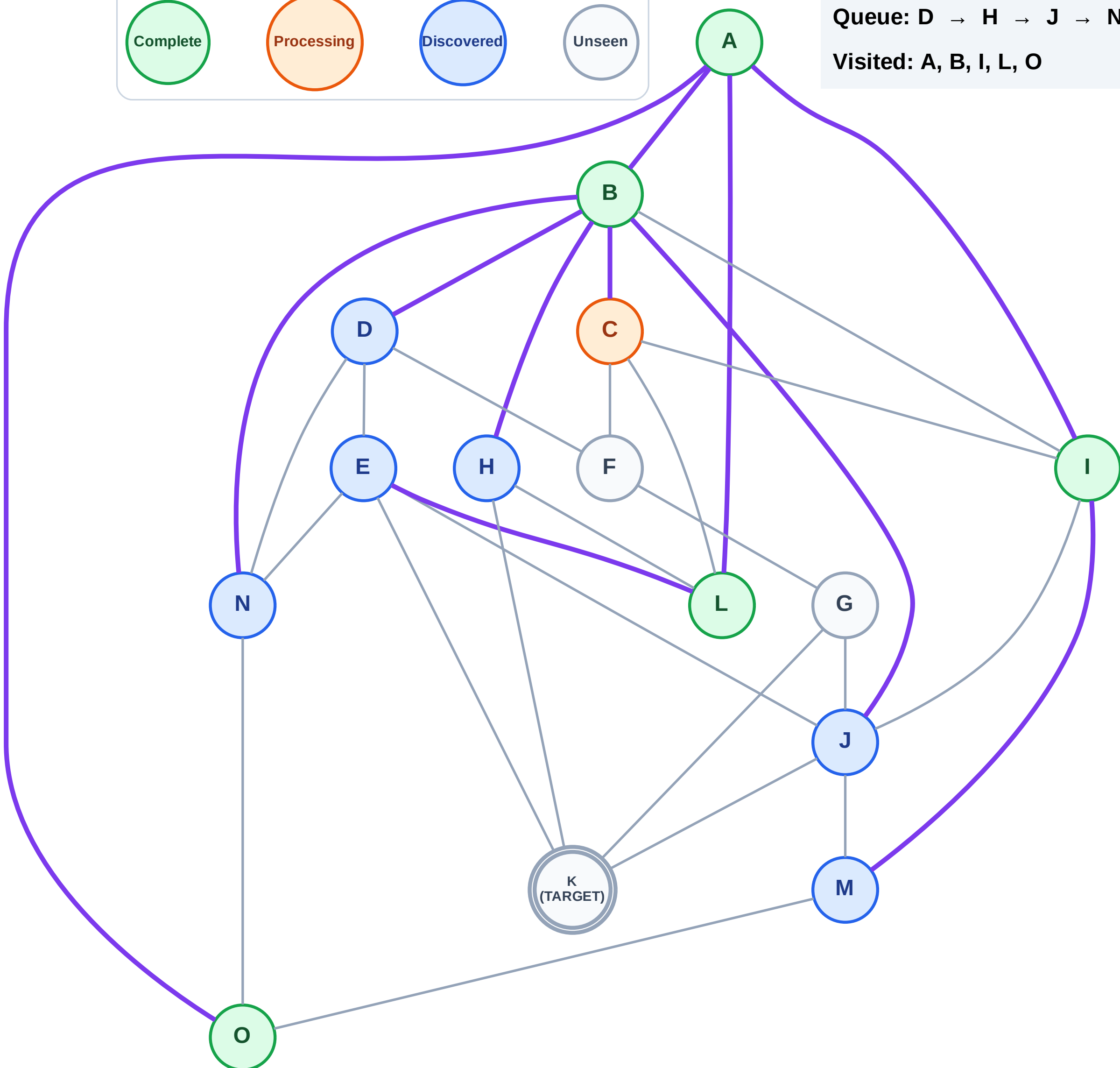
Discovered



Unseen

Queue: D → H → J → N → M → E

Visited: A, B, I, L, O



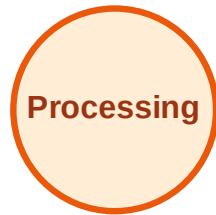
BFS Traversal

Step 13: Discover F from C

Legend



Complete



Processing



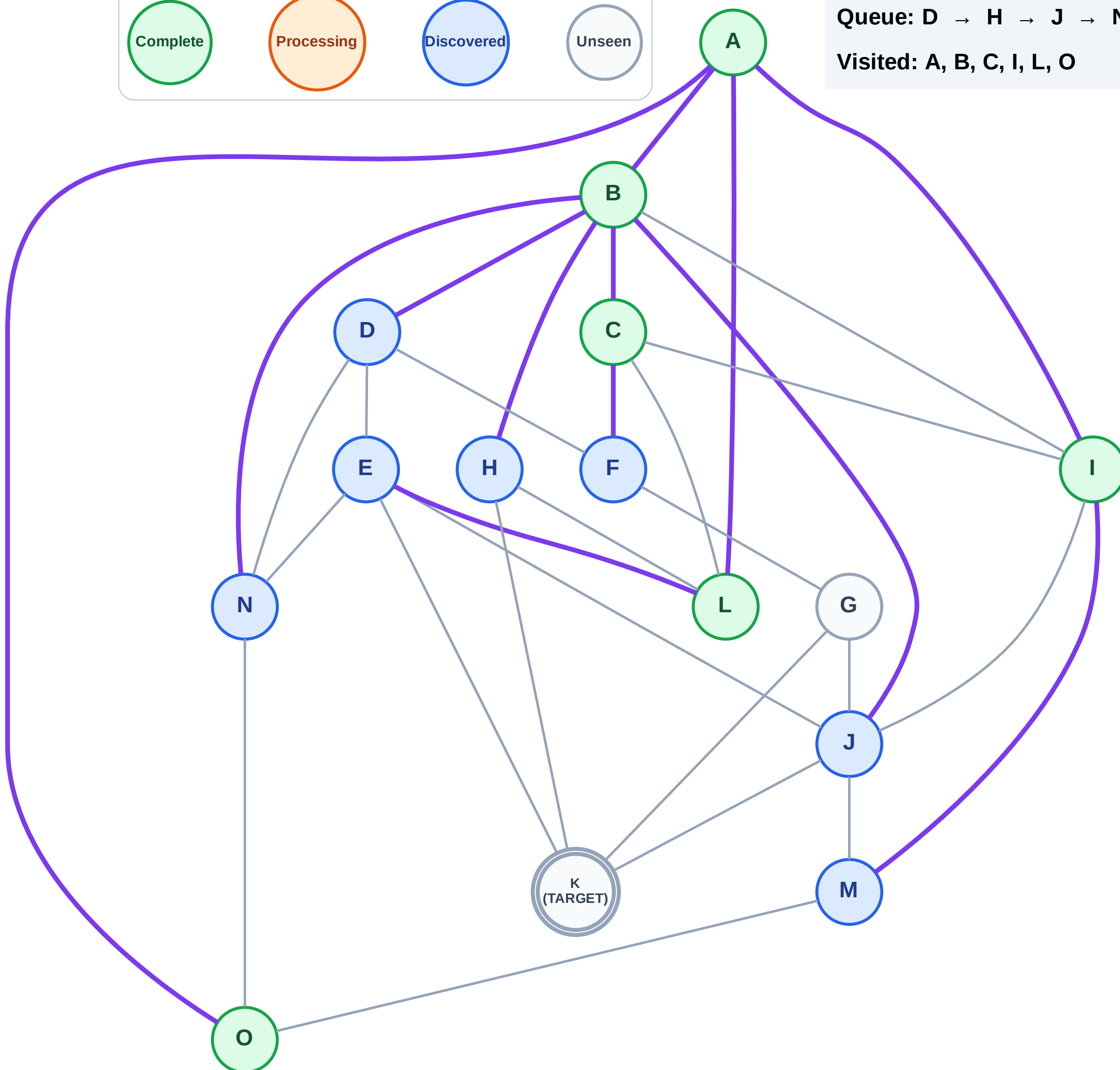
Discovered



Unseen

Queue: D → H → J → N → M → E → F

Visited: A, B, C, I, L, O



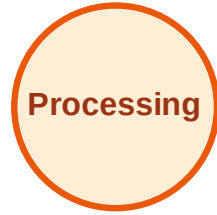
BFS Traversal

Step 14: Process node D

Legend



Complete



Processing



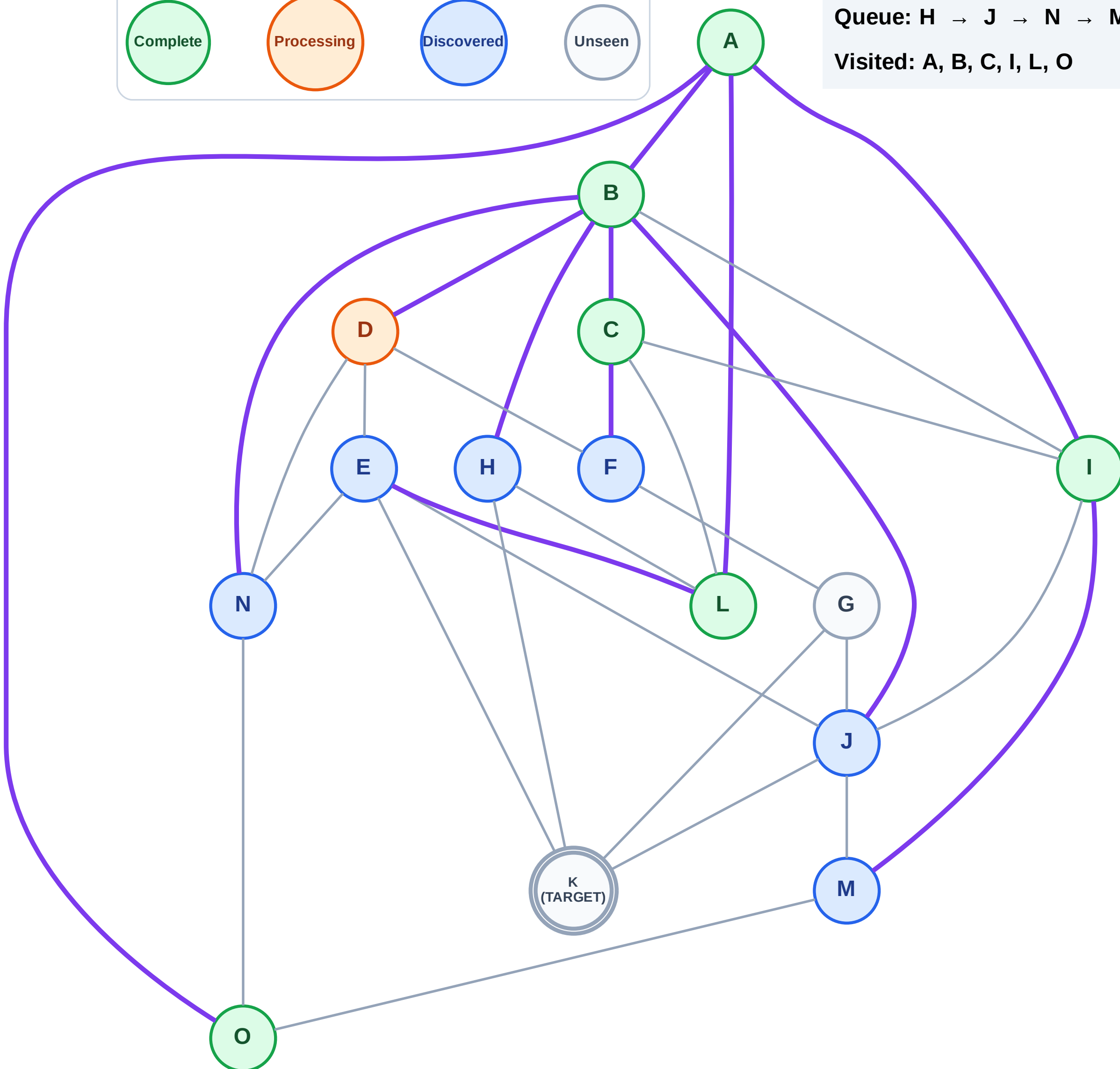
Discovered



Unseen

Queue: H → J → N → M → E → F

Visited: A, B, C, I, L, O



BFS Traversal

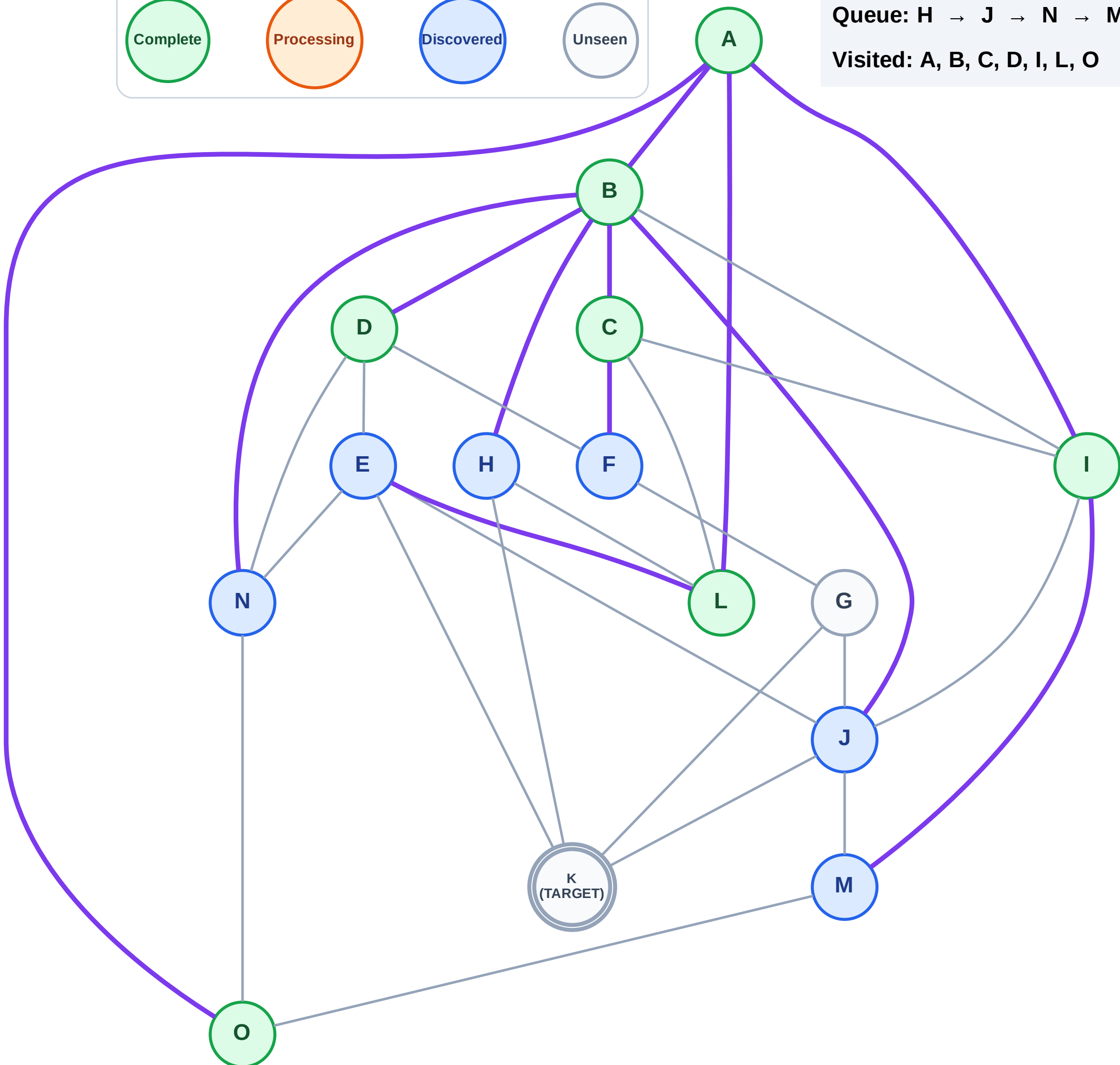
Step 15: No new neighbors from D

Legend



Queue: H → J → N → M → E → F

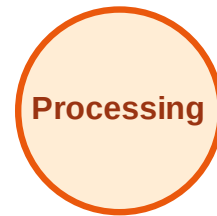
Visited: A, B, C, D, I, L, O



BFS Traversal

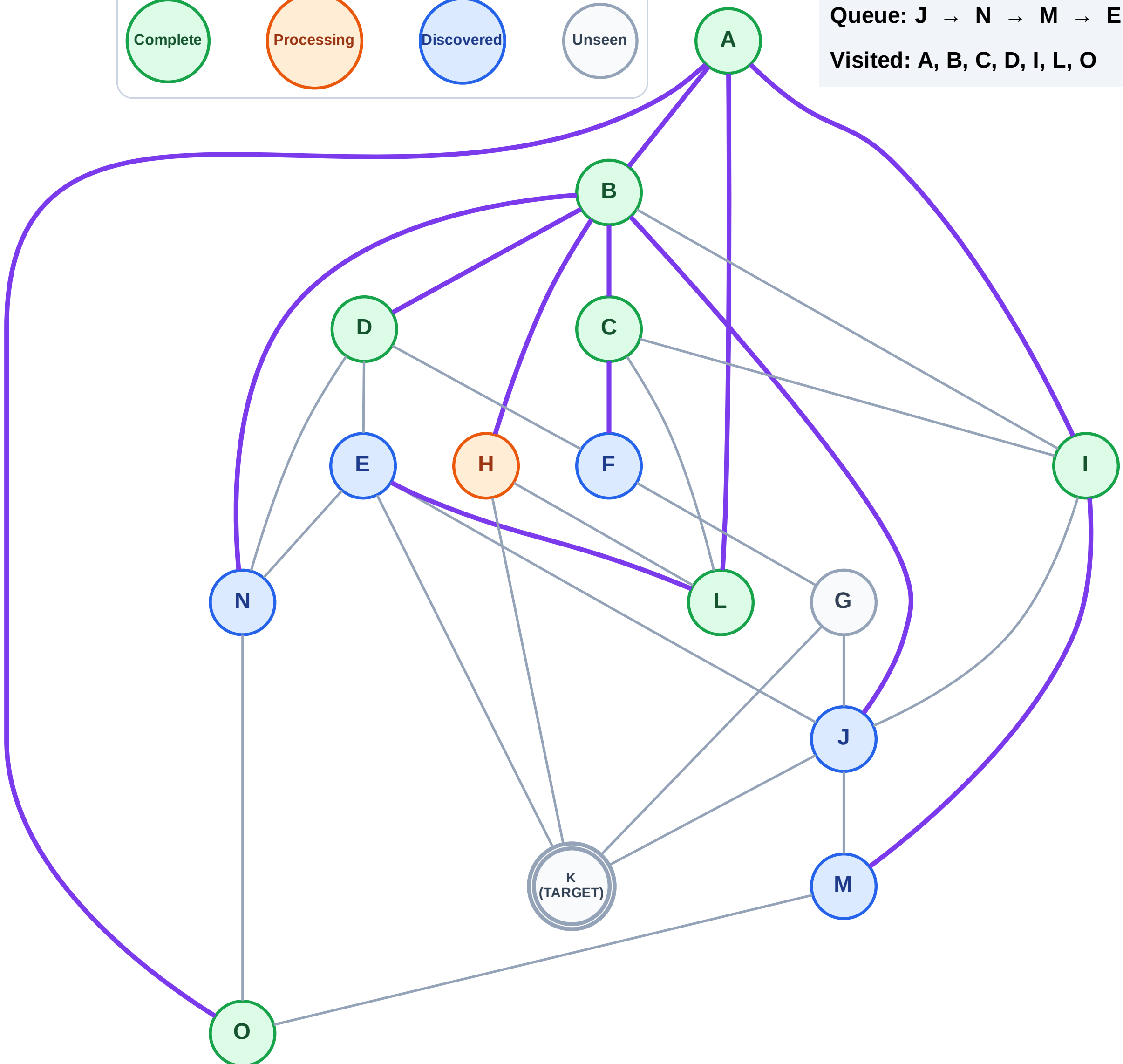
Step 16: Process node H

Legend



Queue: J → N → M → E → F

Visited: A, B, C, D, I, L, O



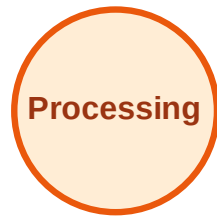
BFS Traversal

Step 17: Discover K from H

Legend



Complete



Processing



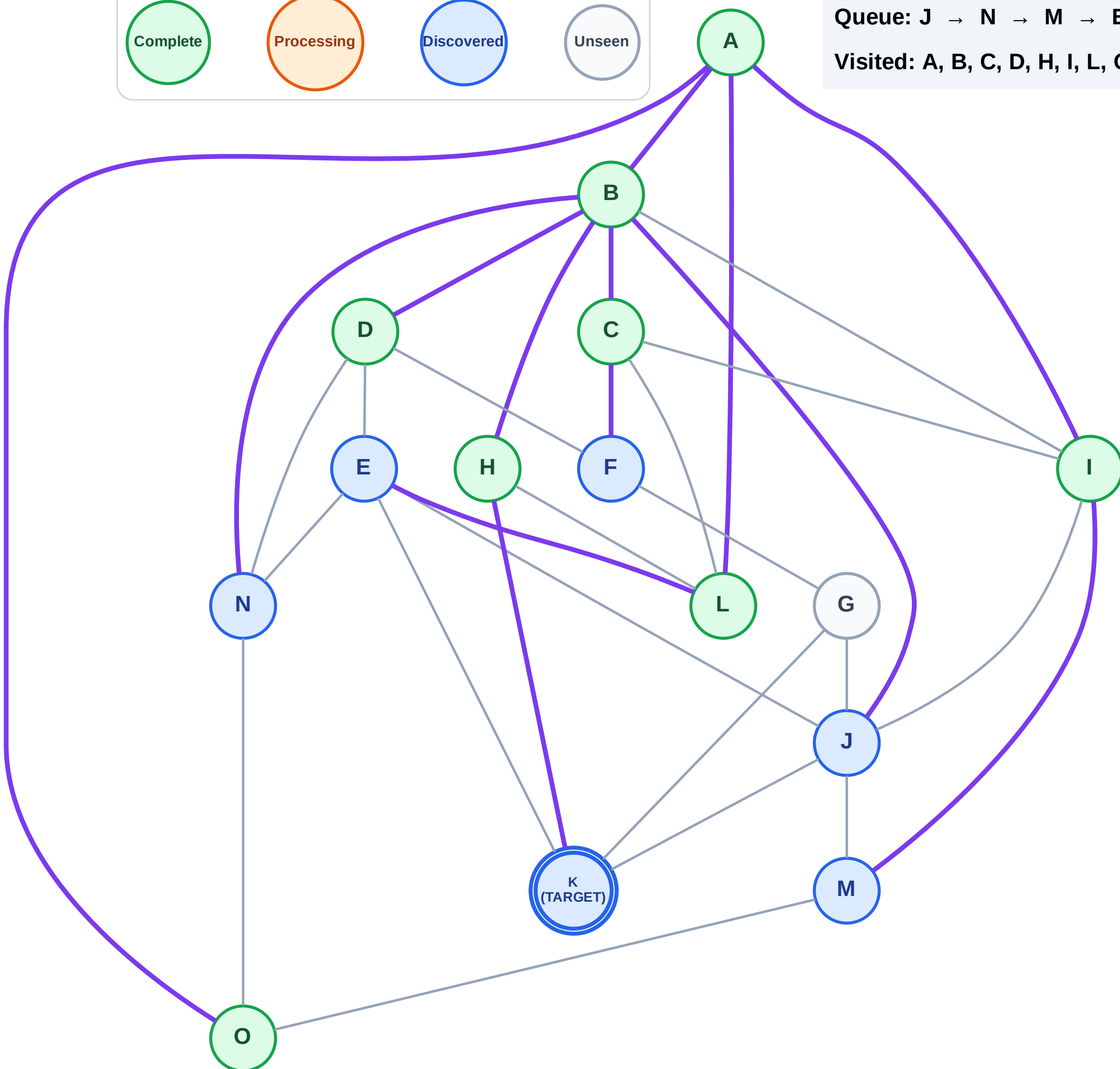
Discovered



Unseen

Queue: J → N → M → E → F → K

Visited: A, B, C, D, H, I, L, O



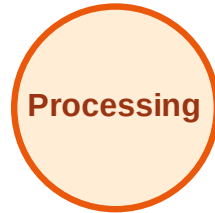
BFS Traversal

Step 18: Process node J

Legend



Complete



Processing



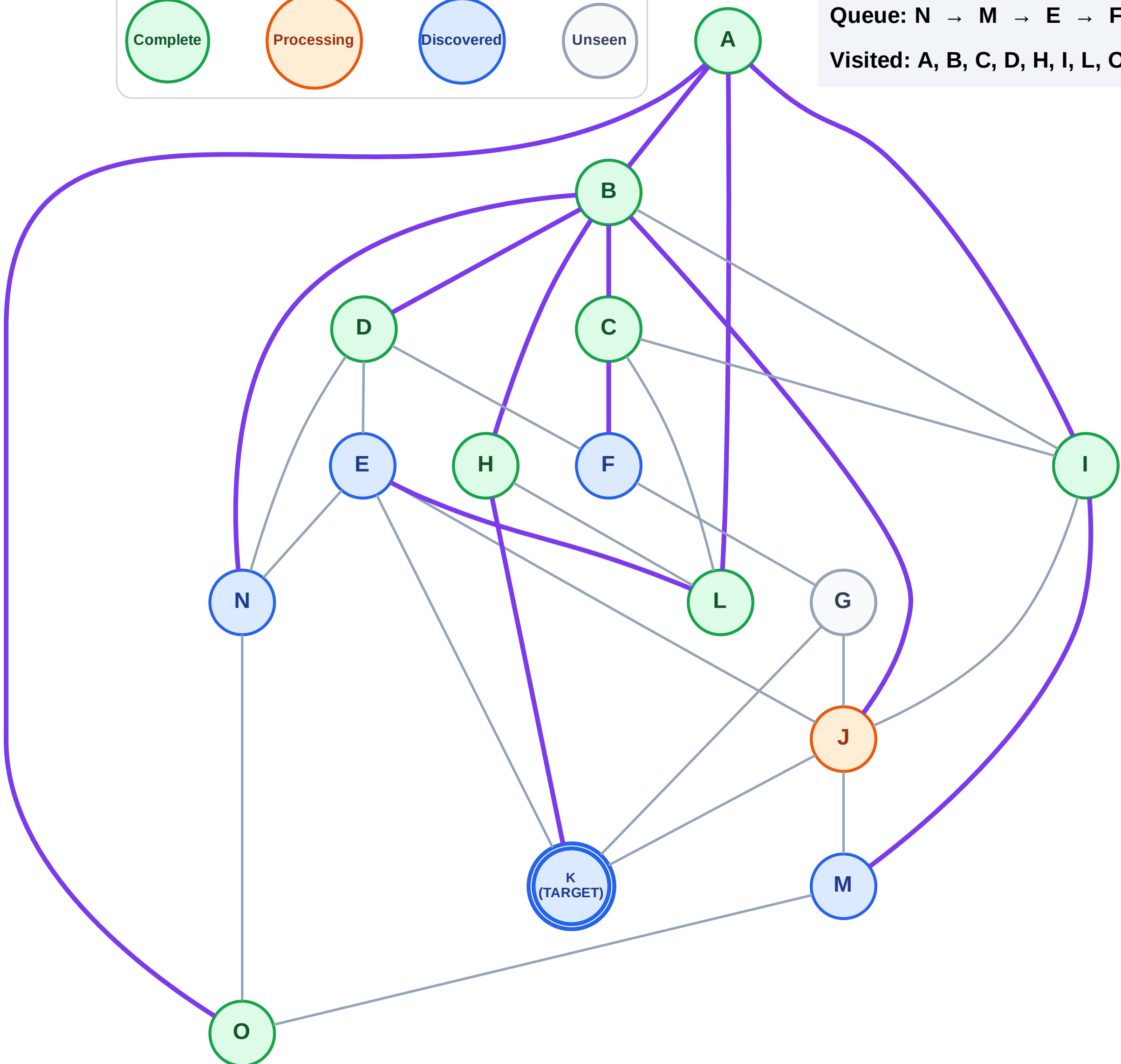
Discovered



Unseen

Queue: N → M → E → F → K

Visited: A, B, C, D, H, I, L, O



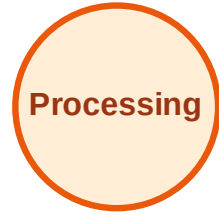
BFS Traversal

Step 19: Discover G from J

Legend



Complete



Processing



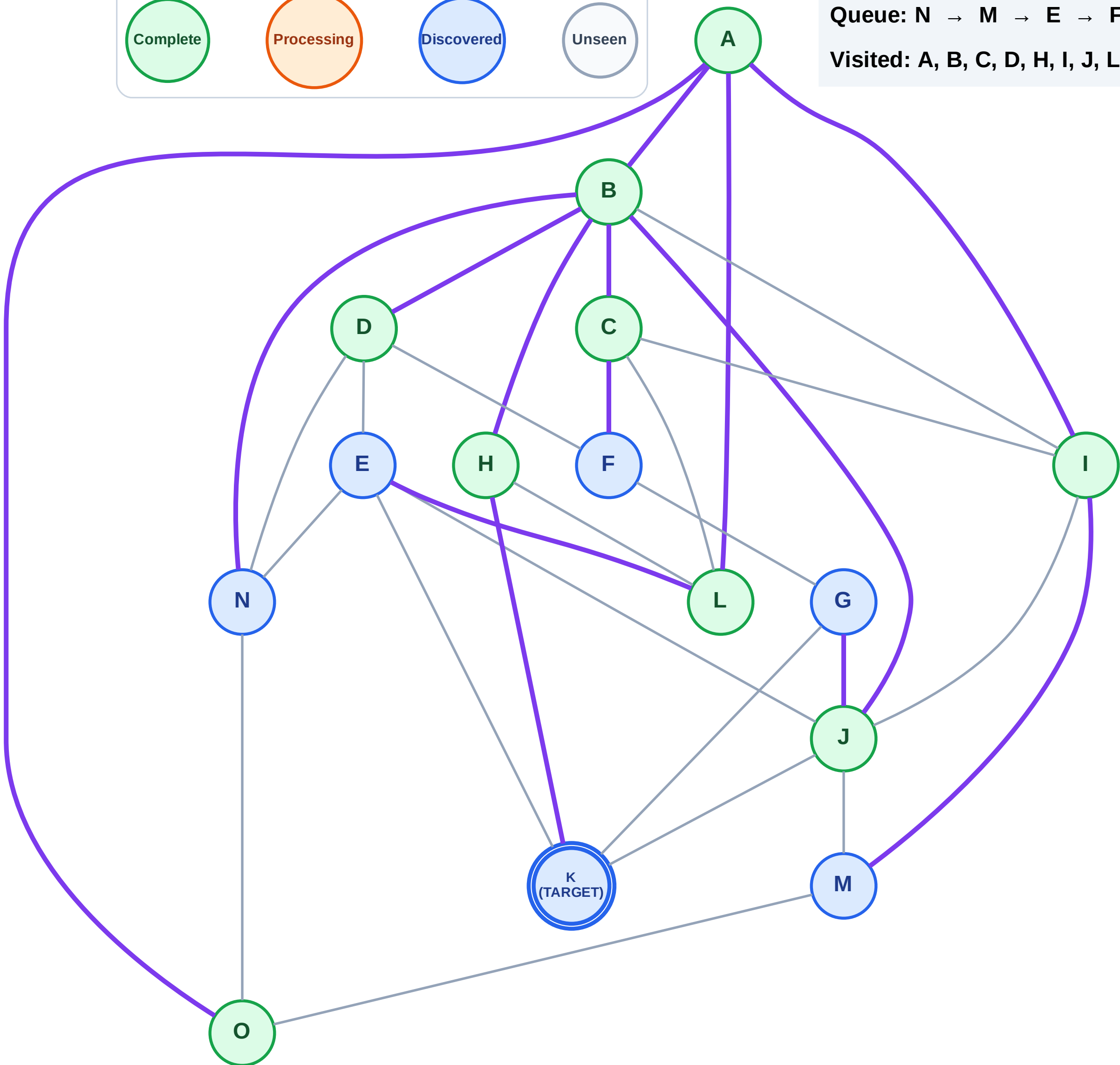
Discovered



Unseen

Queue: N → M → E → F → K → G

Visited: A, B, C, D, H, I, J, L, O



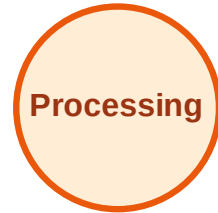
BFS Traversal

Step 20: Process node N

Legend



Complete



Processing



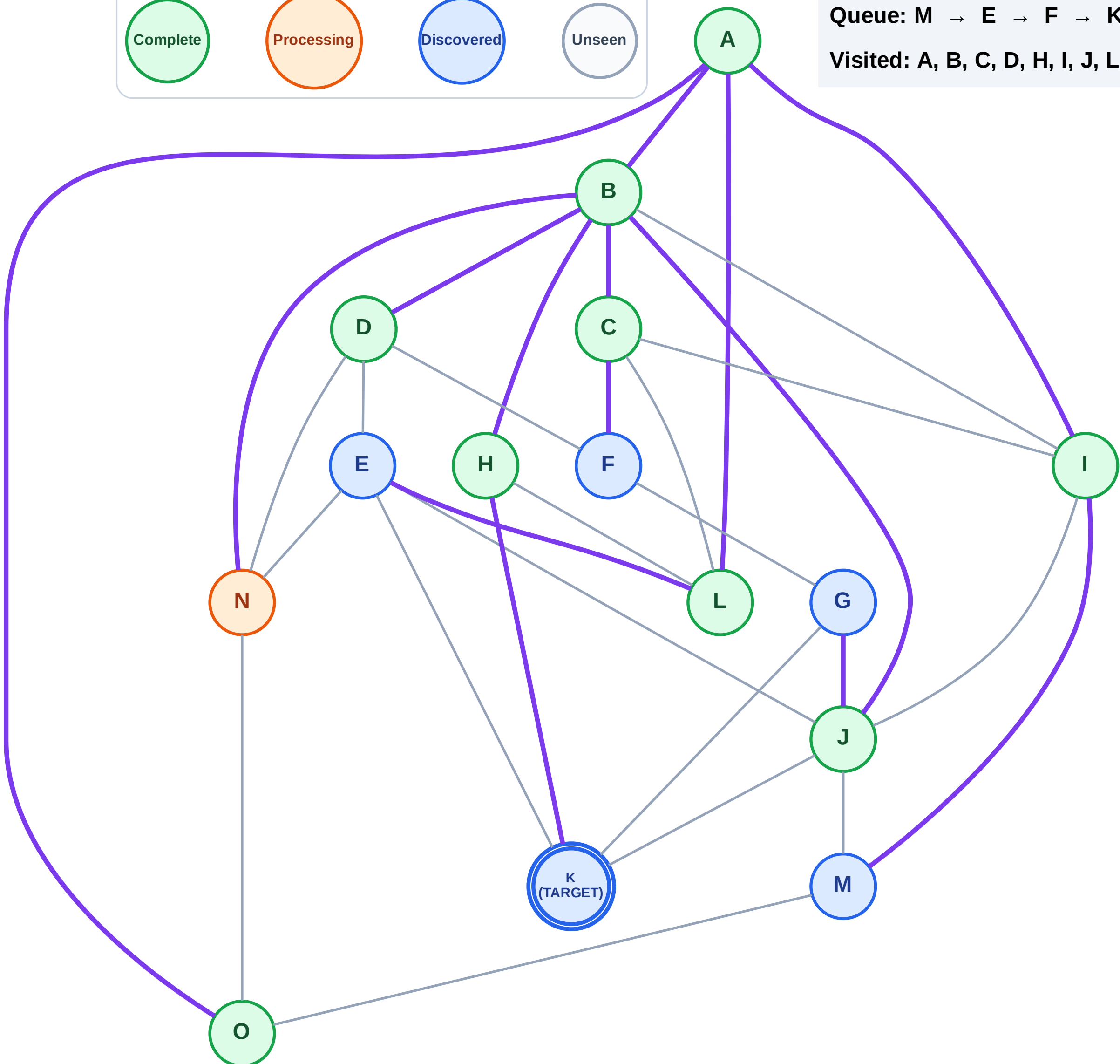
Discovered



Unseen

Queue: M → E → F → K → G

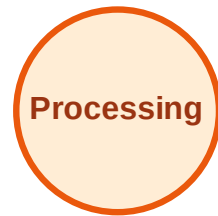
Visited: A, B, C, D, H, I, J, L, O



BFS Traversal

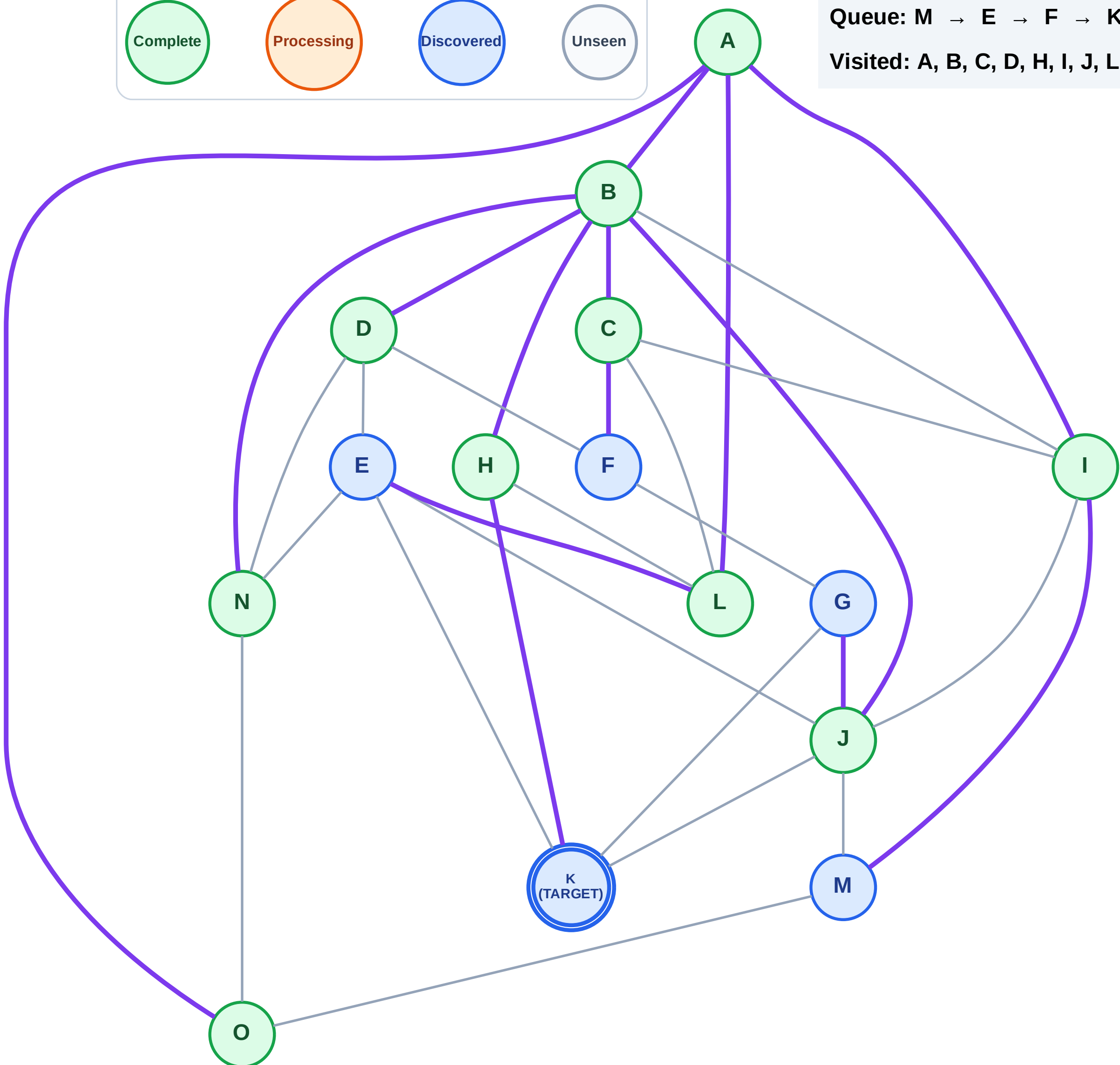
Step 21: No new neighbors from N

Legend



Queue: M → E → F → K → G

Visited: A, B, C, D, H, I, J, L, N, O



BFS Traversal

Step 22: Process node M

Legend

Complete

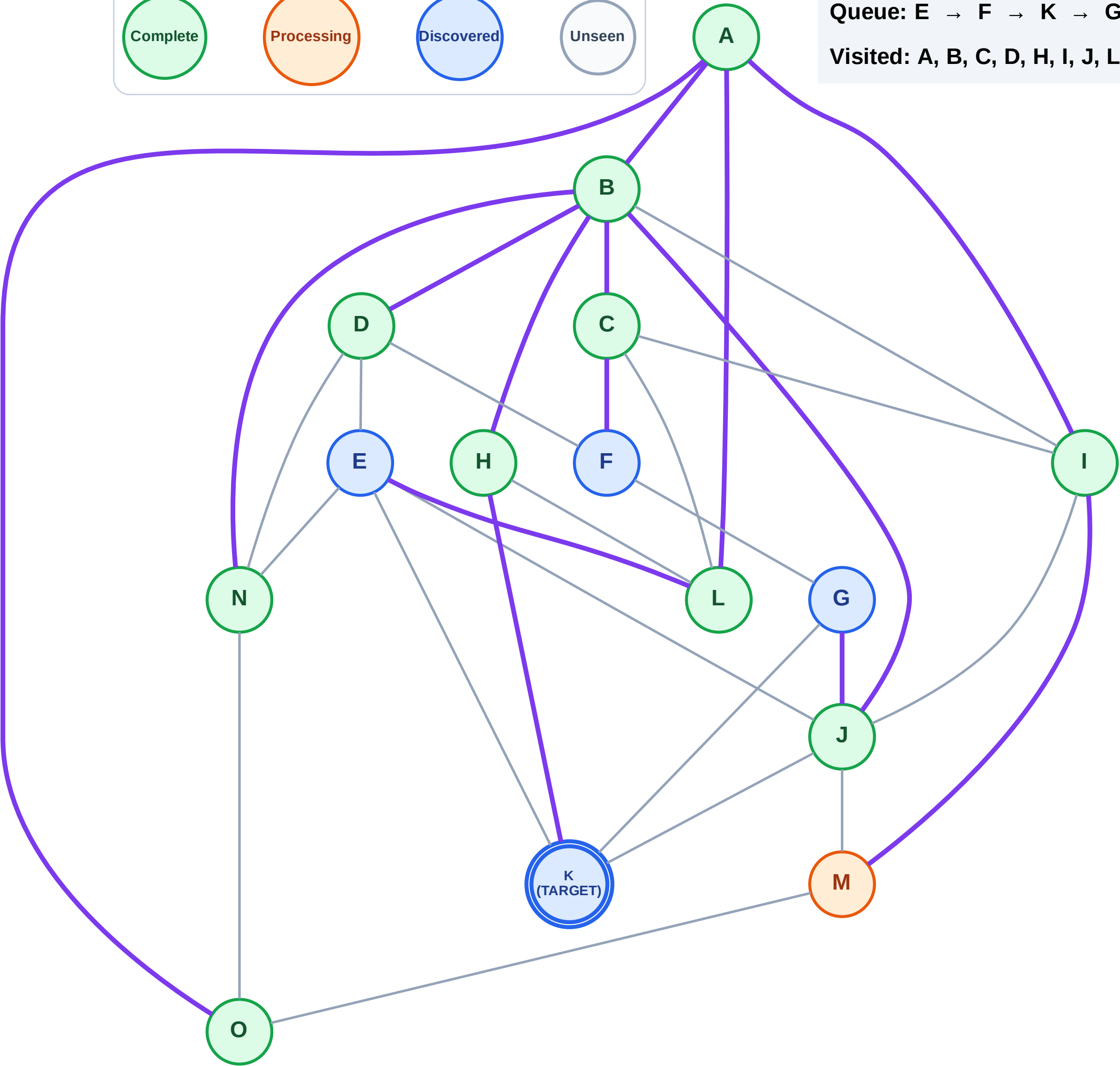
Processing

Discovered

Unseen

Queue: E → F → K → G

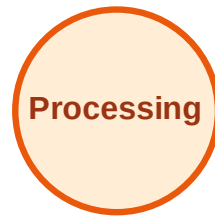
Visited: A, B, C, D, H, I, J, L, N, O



BFS Traversal

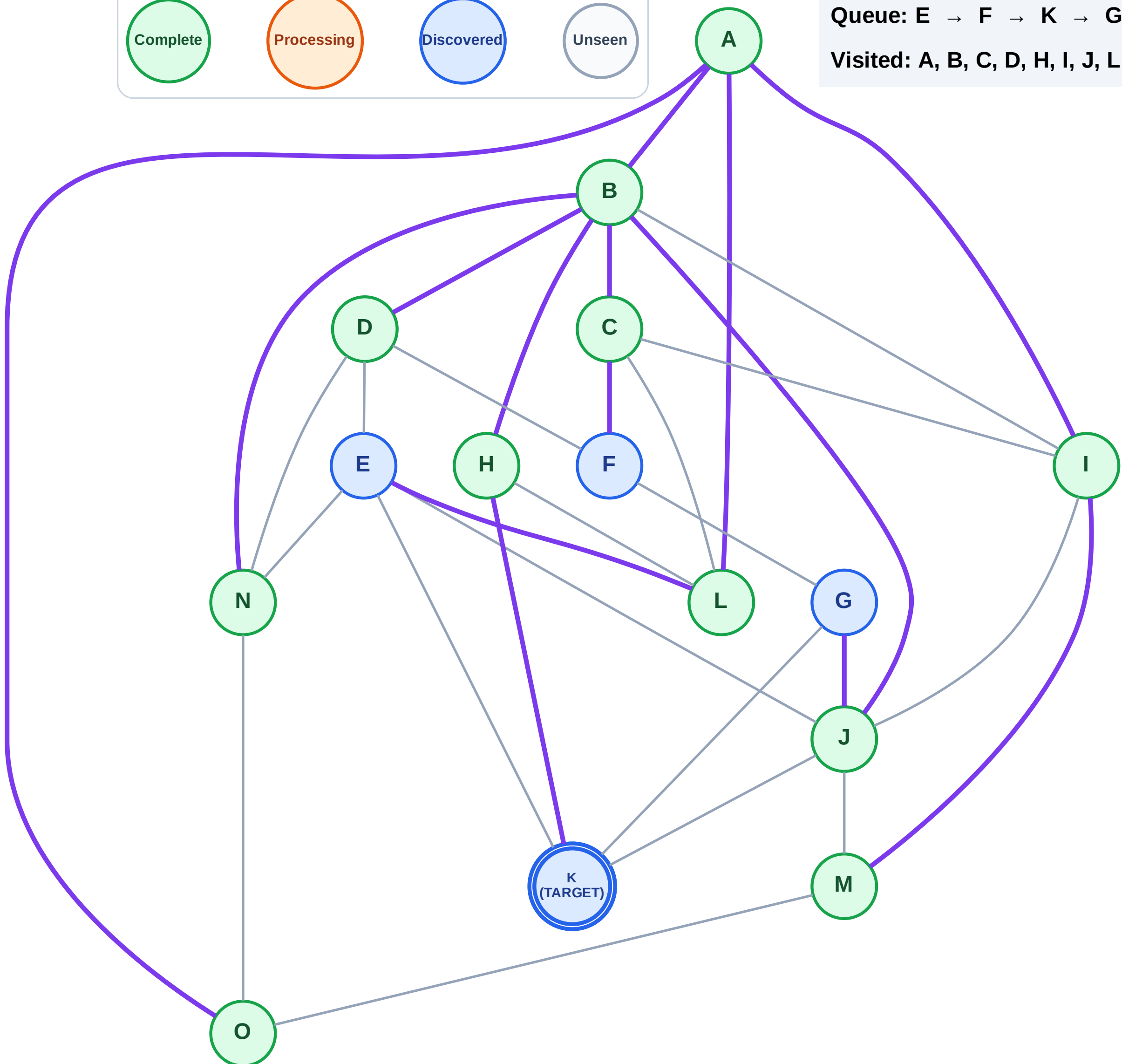
Step 23: No new neighbors from M

Legend



Queue: E → F → K → G

Visited: A, B, C, D, H, I, J, L, M, N, O



BFS Traversal

Step 24: Process node E

Legend

Complete

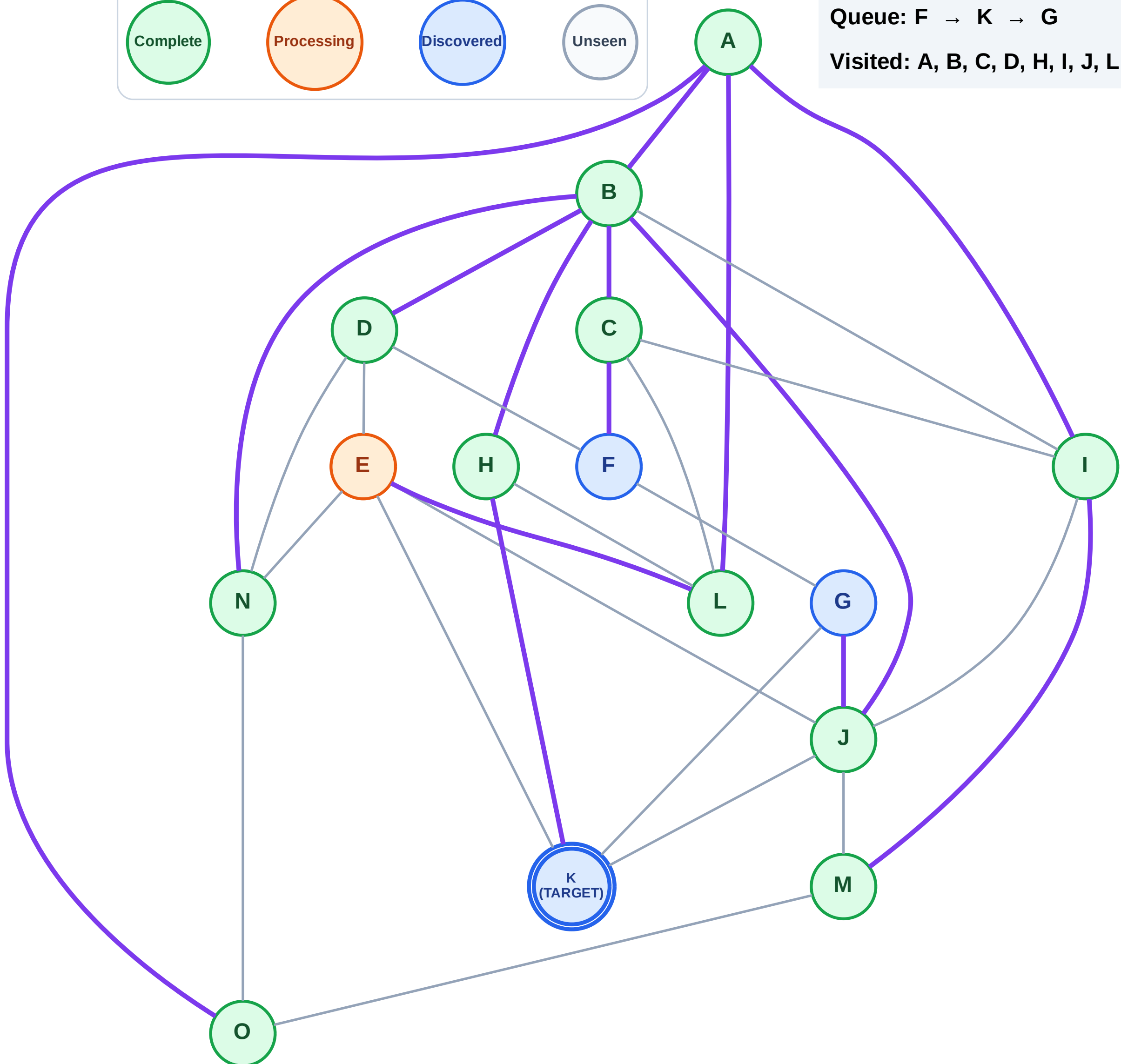
Processing

Discovered

Unseen

Queue: F → K → G

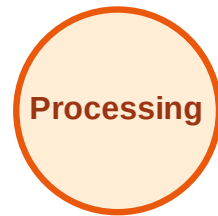
Visited: A, B, C, D, H, I, J, L, M, N, O



BFS Traversal

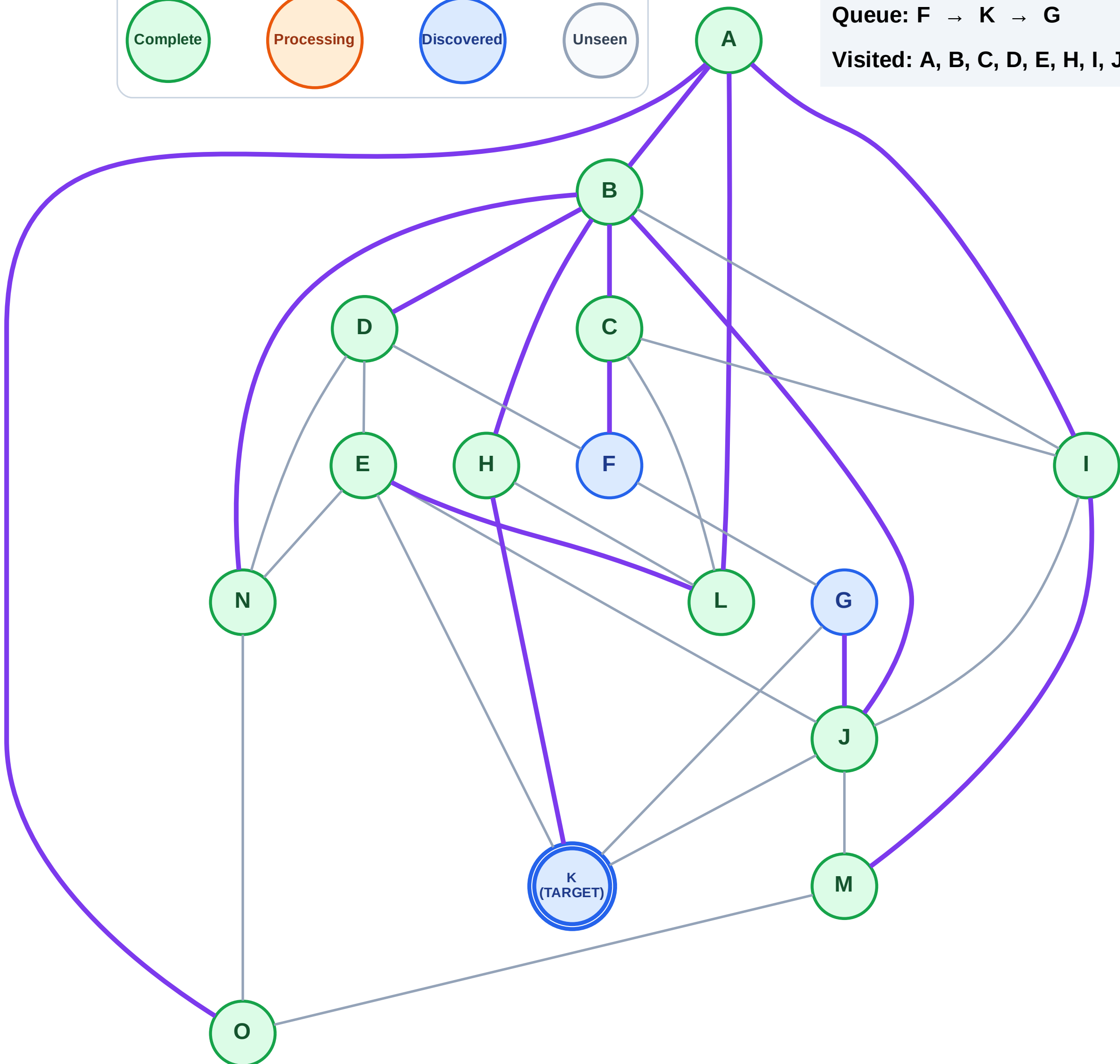
Step 25: No new neighbors from E

Legend



Queue: F → K → G

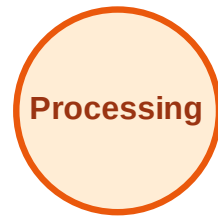
Visited: A, B, C, D, E, H, I, J, L, M, N, O



BFS Traversal

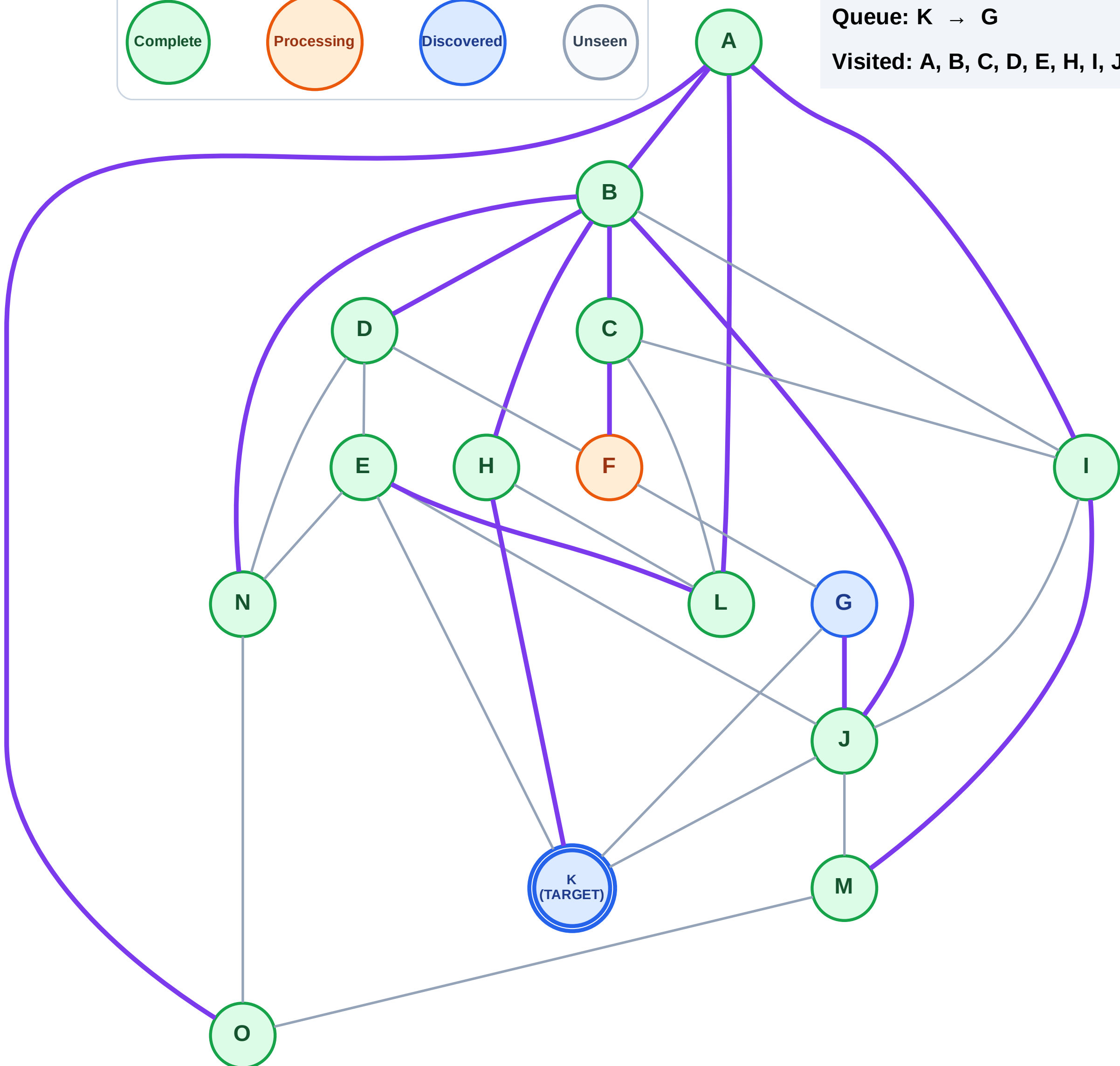
Step 26: Process node F

Legend



Queue: K → G

Visited: A, B, C, D, E, H, I, J, L, M, N, O



BFS Traversal

Step 27: No new neighbors from F

Legend

Complete

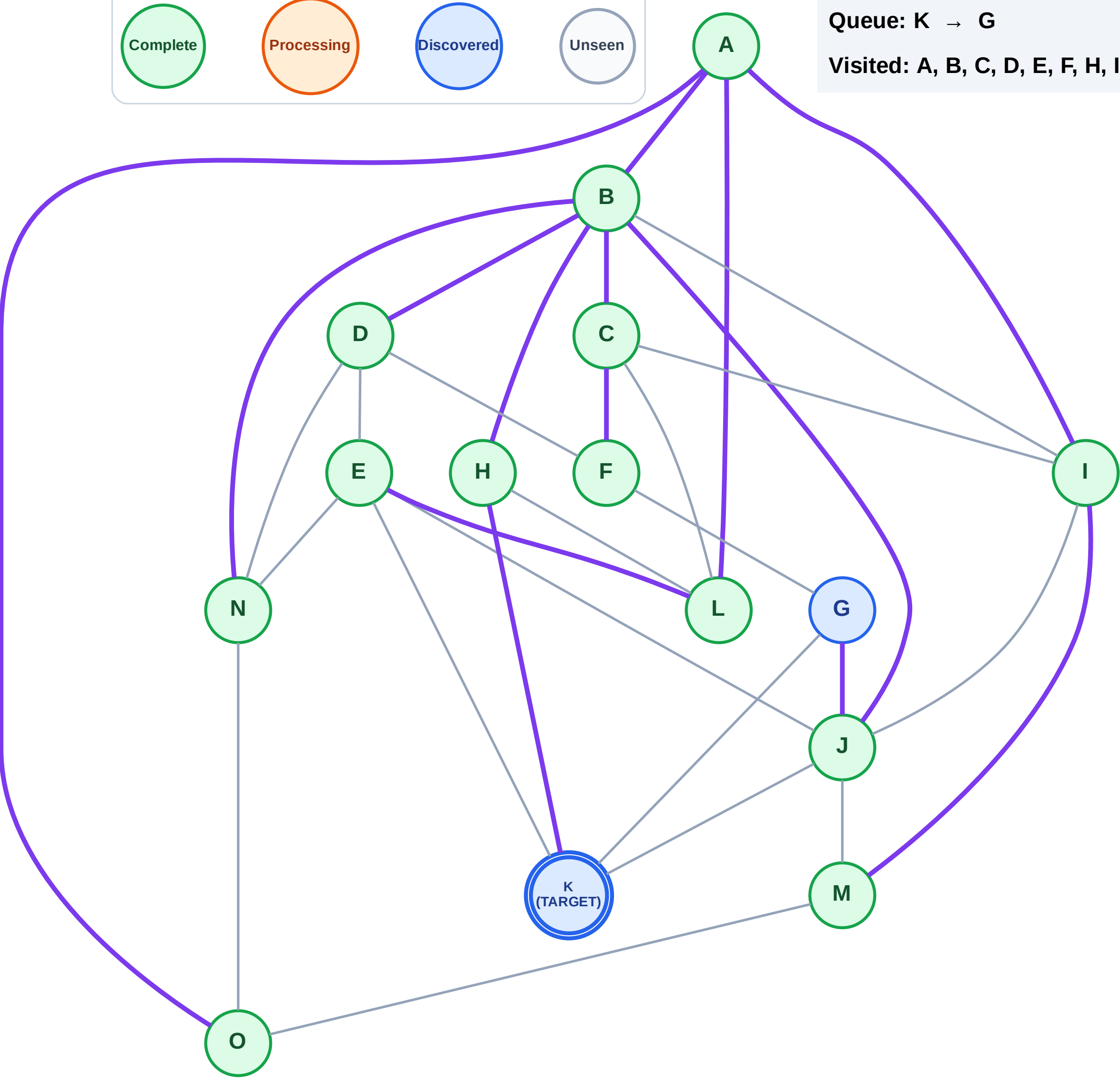
Processing

Discovered

Unseen

Queue: K → G

Visited: A, B, C, D, E, F, H, I, J, L, M, N, O



BFS Traversal

Step 28: Process node K

Legend

Complete

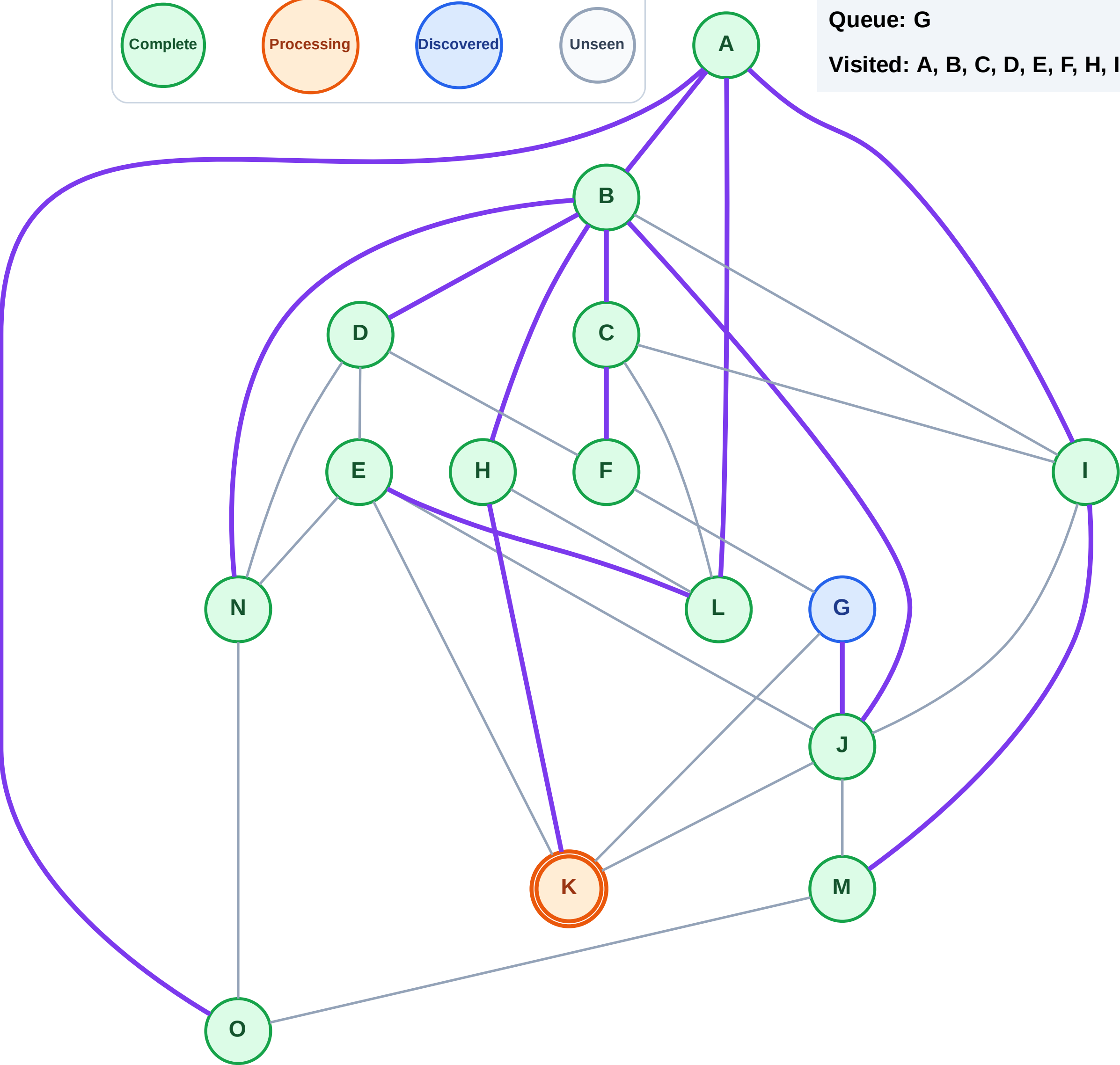
Processing

Discovered

Unseen

Queue: G

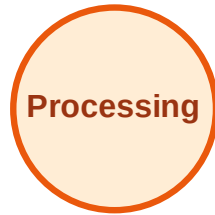
Visited: A, B, C, D, E, F, H, I, J, L, M, N, O



BFS Traversal

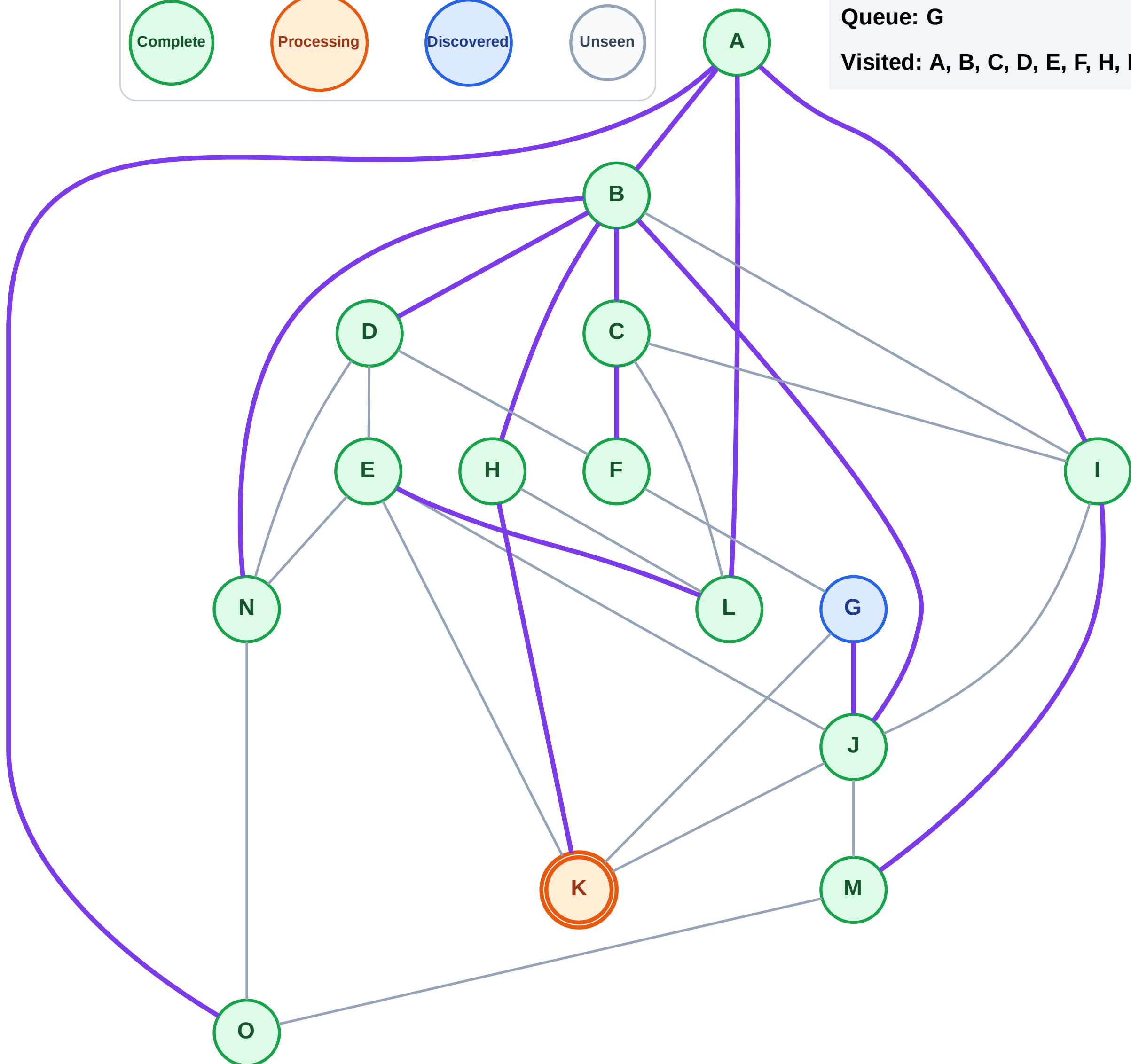
Step 29: TARGET NODE K REACHED! Path (3 hops): A → B → H → K

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J, K, L, M, N, O



DFS Traversal

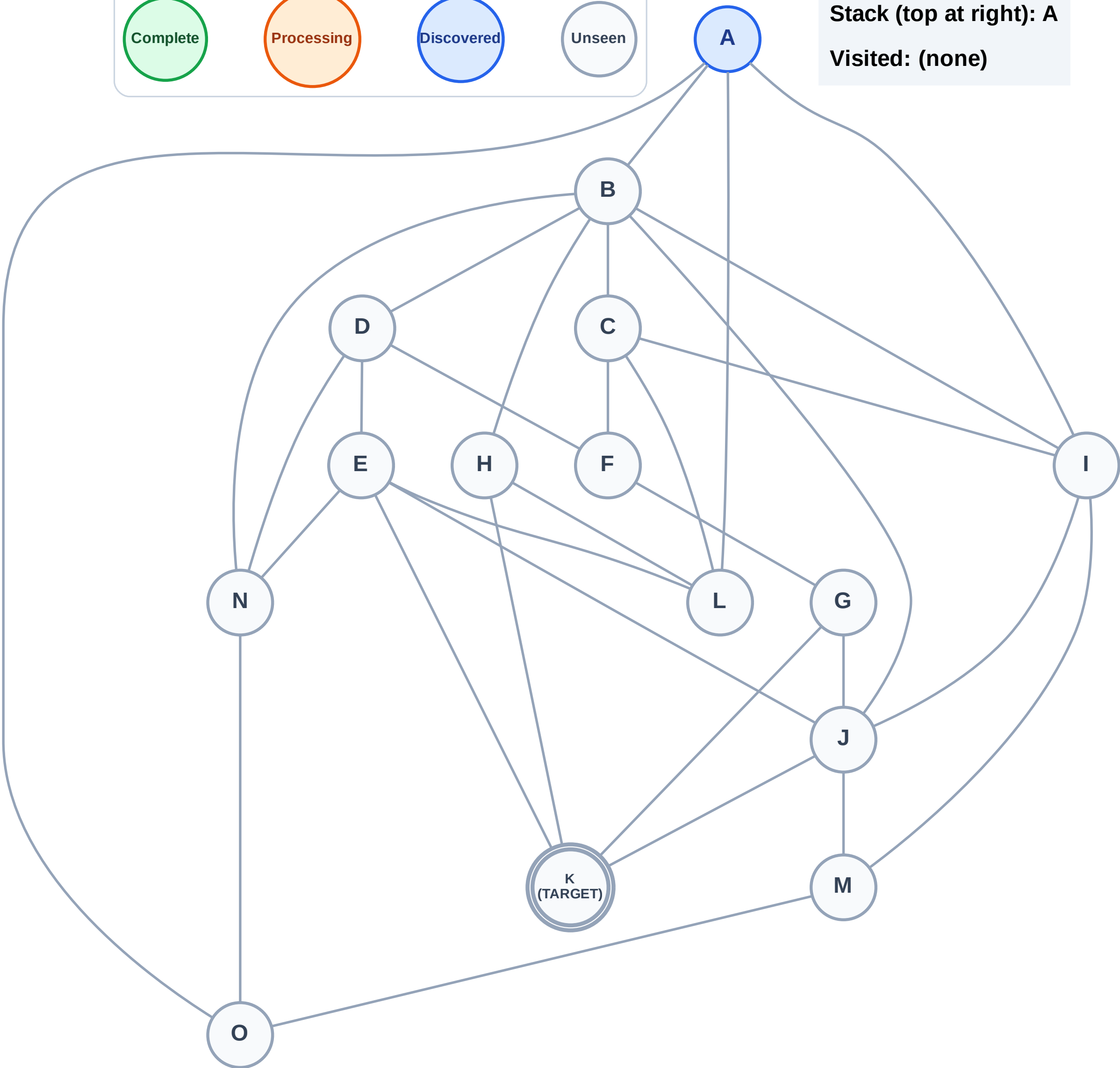
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



Step 2: Process node A

Step 2: Process node A

Legend

Complete Processing Discovered Unseen

Complete

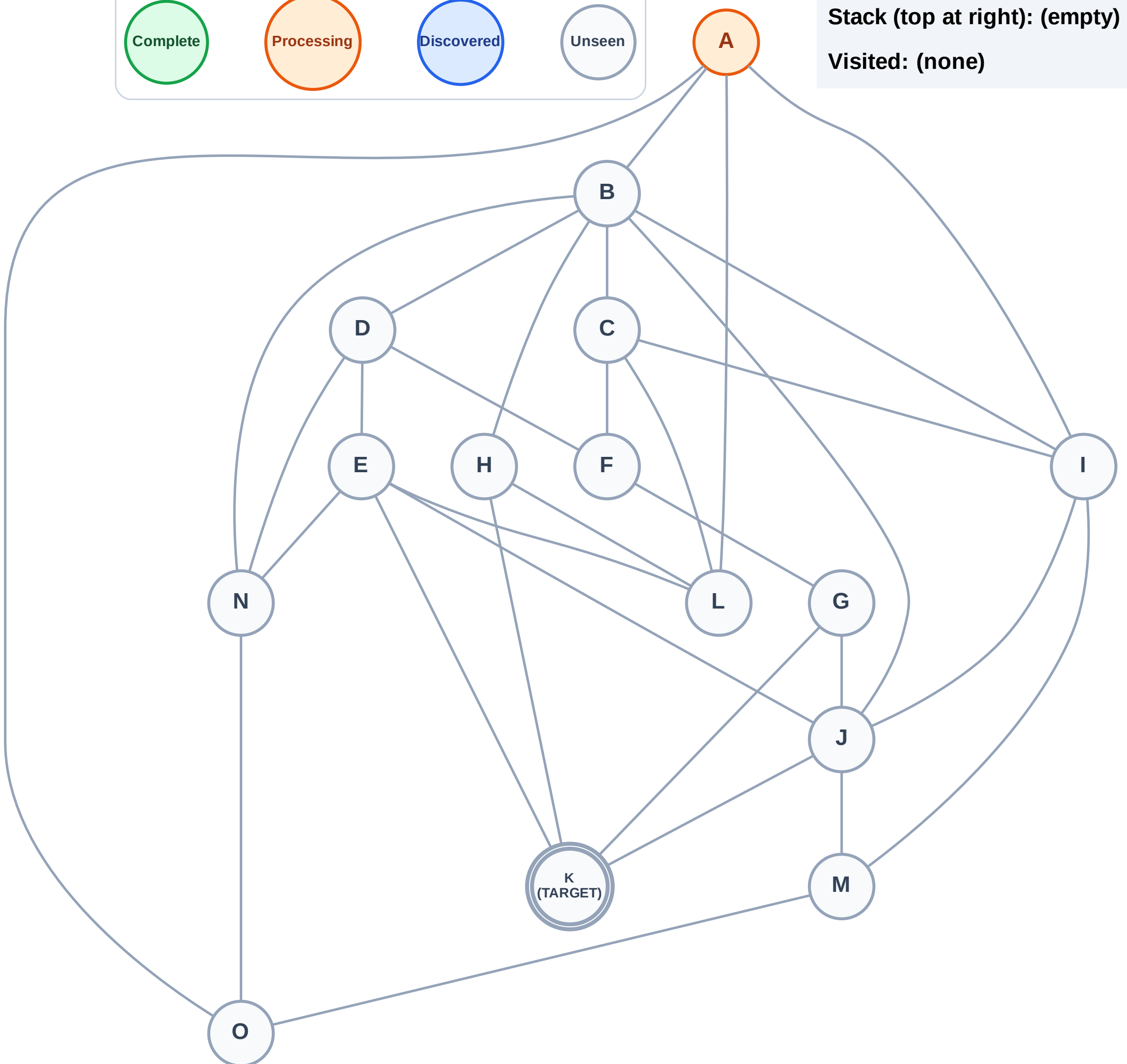
Processing

Discovered

Unseen

Stack (top at right): (empty)

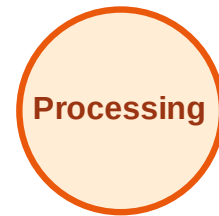
Visited: (none)



DFS Traversal

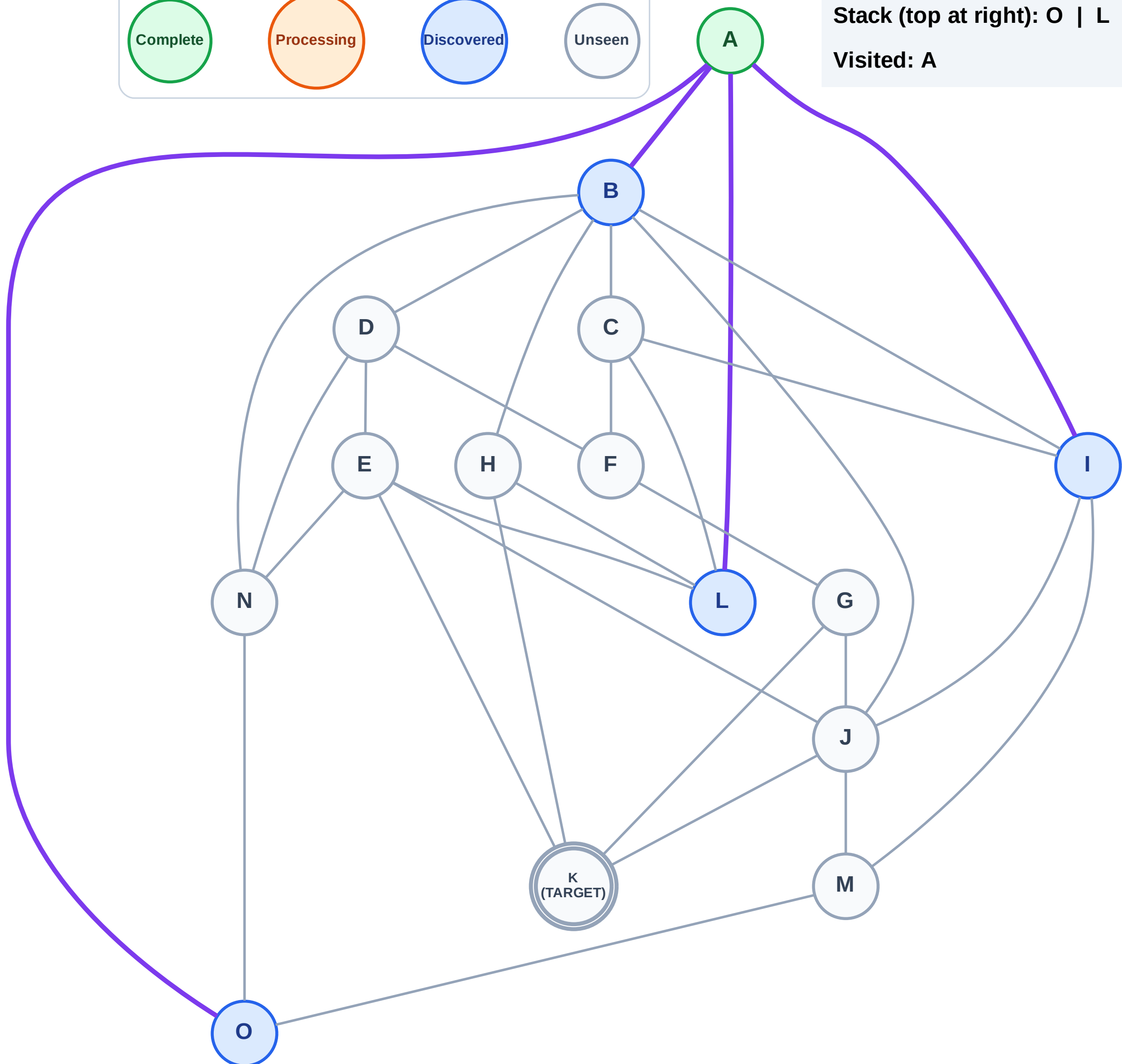
Step 3: Discover O, L, I, B from A

Legend



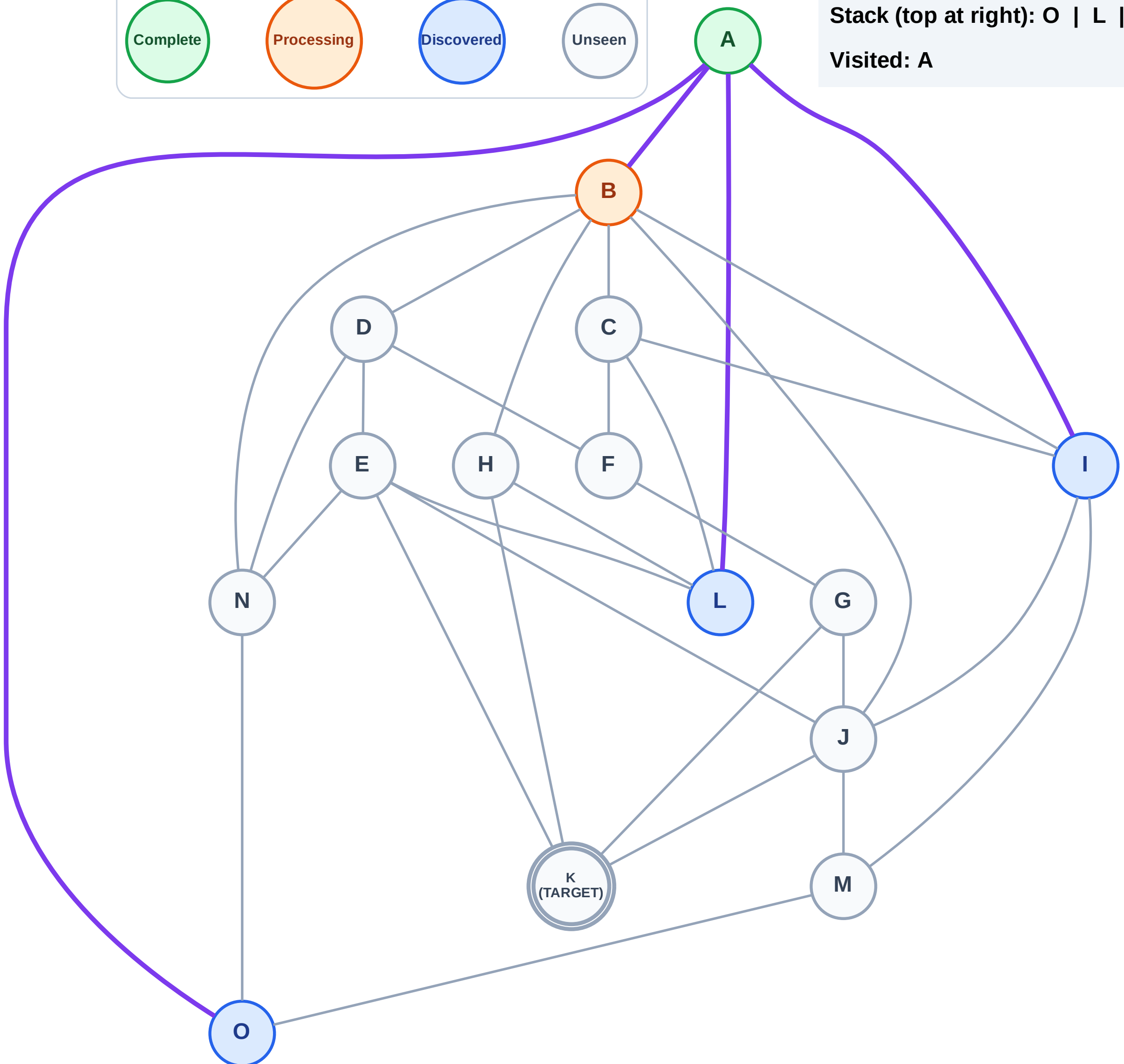
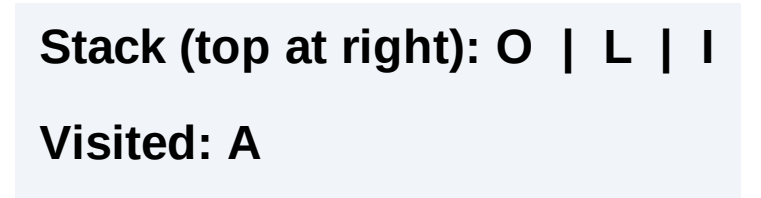
Stack (top at right): O | L | I | B

Visited: A



Step 4: Process node B

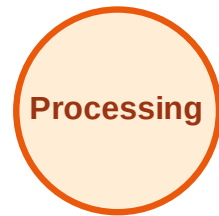
Step 4: Process node B



DFS Traversal

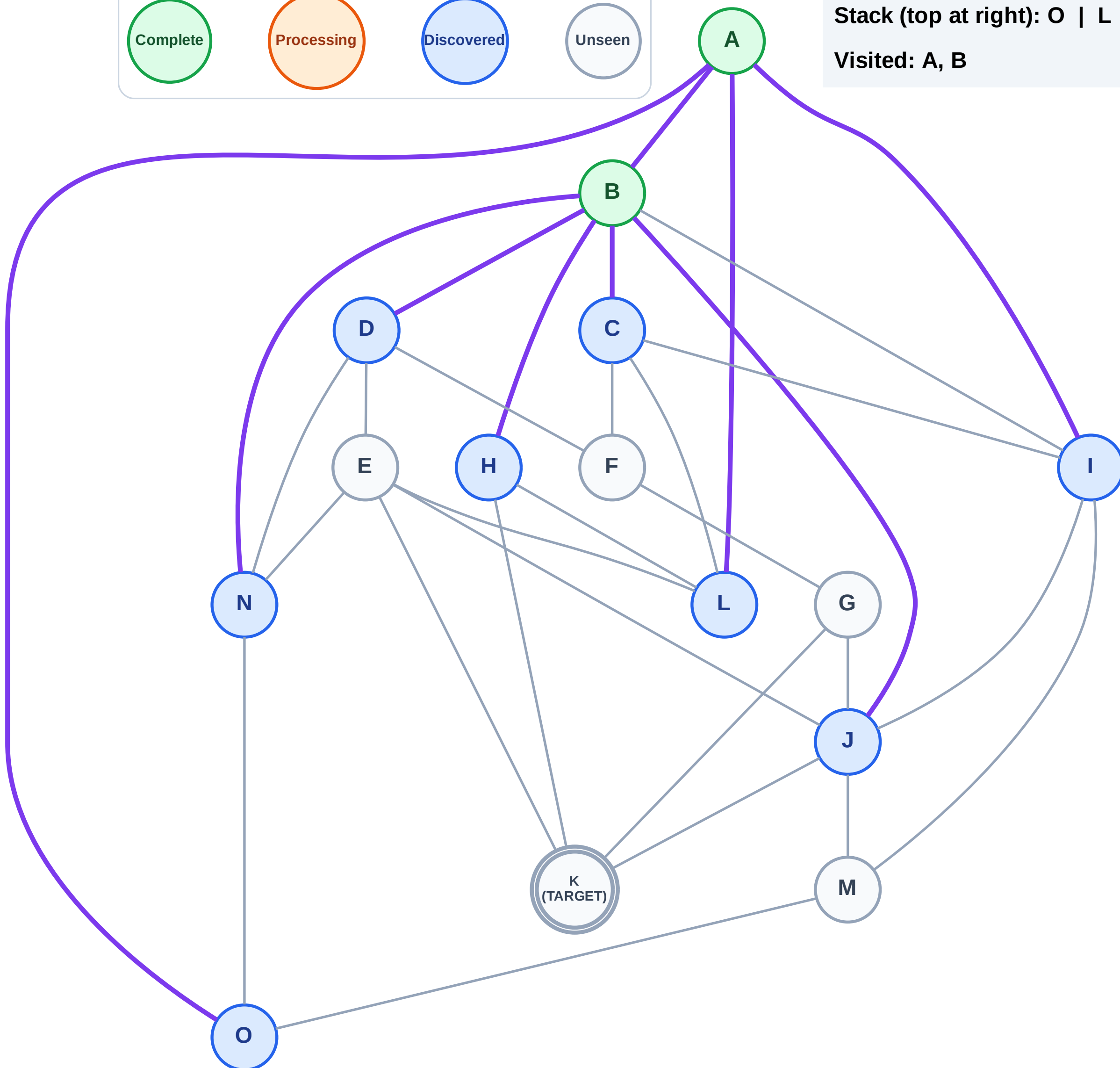
Step 5: Discover N, J, H, D, C from B

Legend



Stack (top at right): O | L | I | N | J | H | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

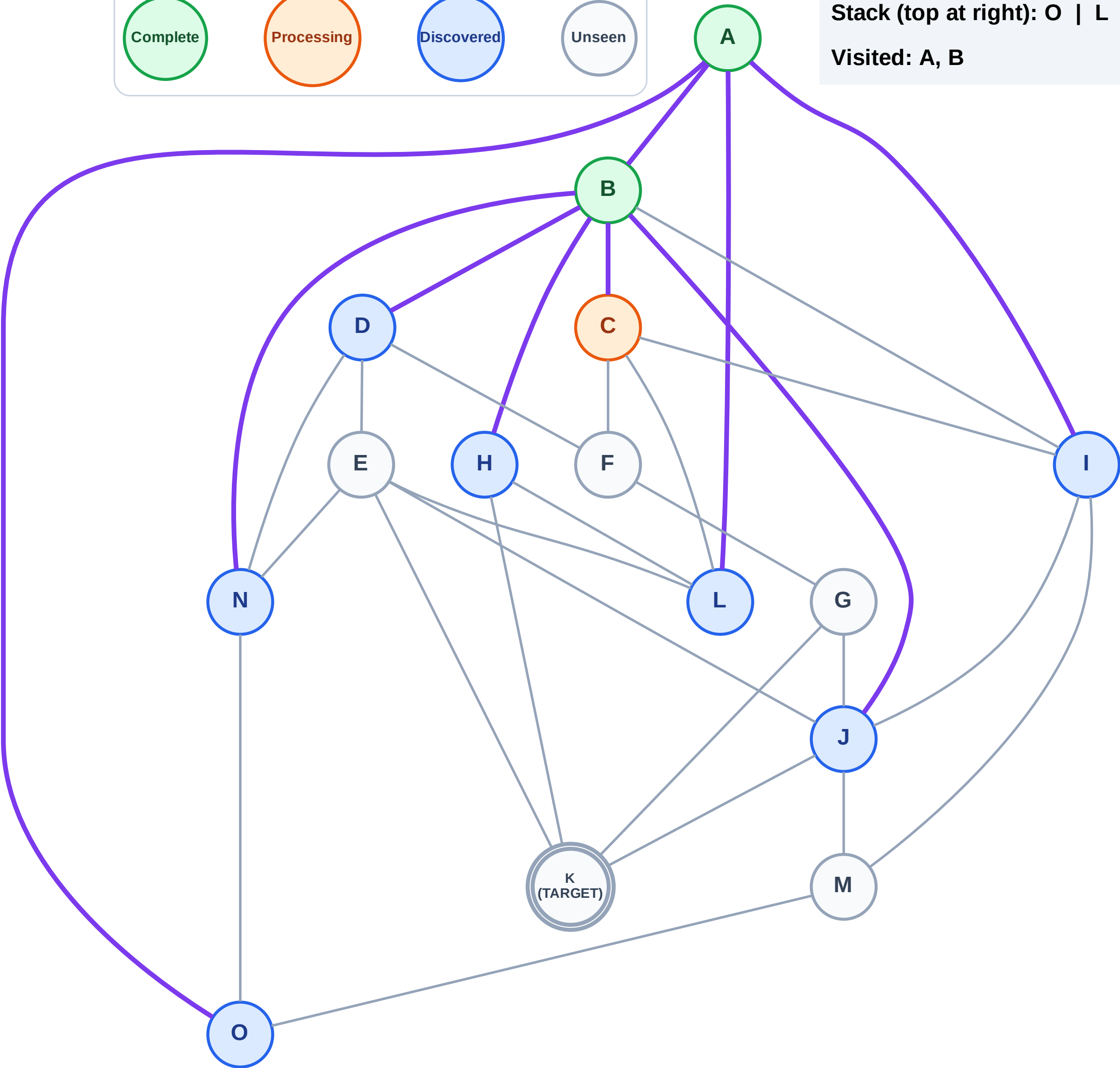
Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

Visited: A, B



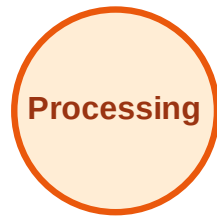
DFS Traversal

Step 7: Discover F from C

Legend



Complete



Processing



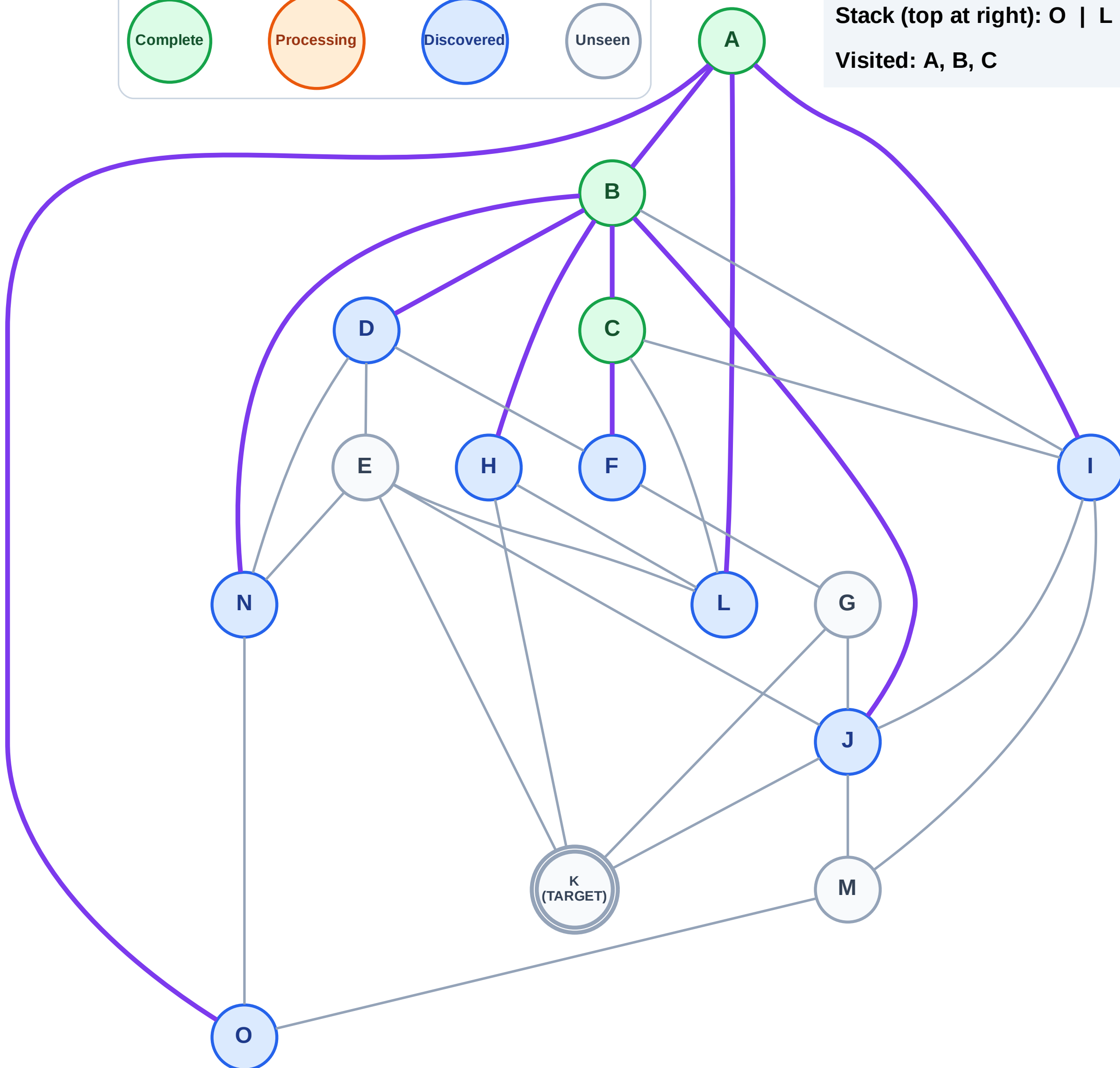
Discovered



Unseen

Stack (top at right): O | L | I | N | J | H | D | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

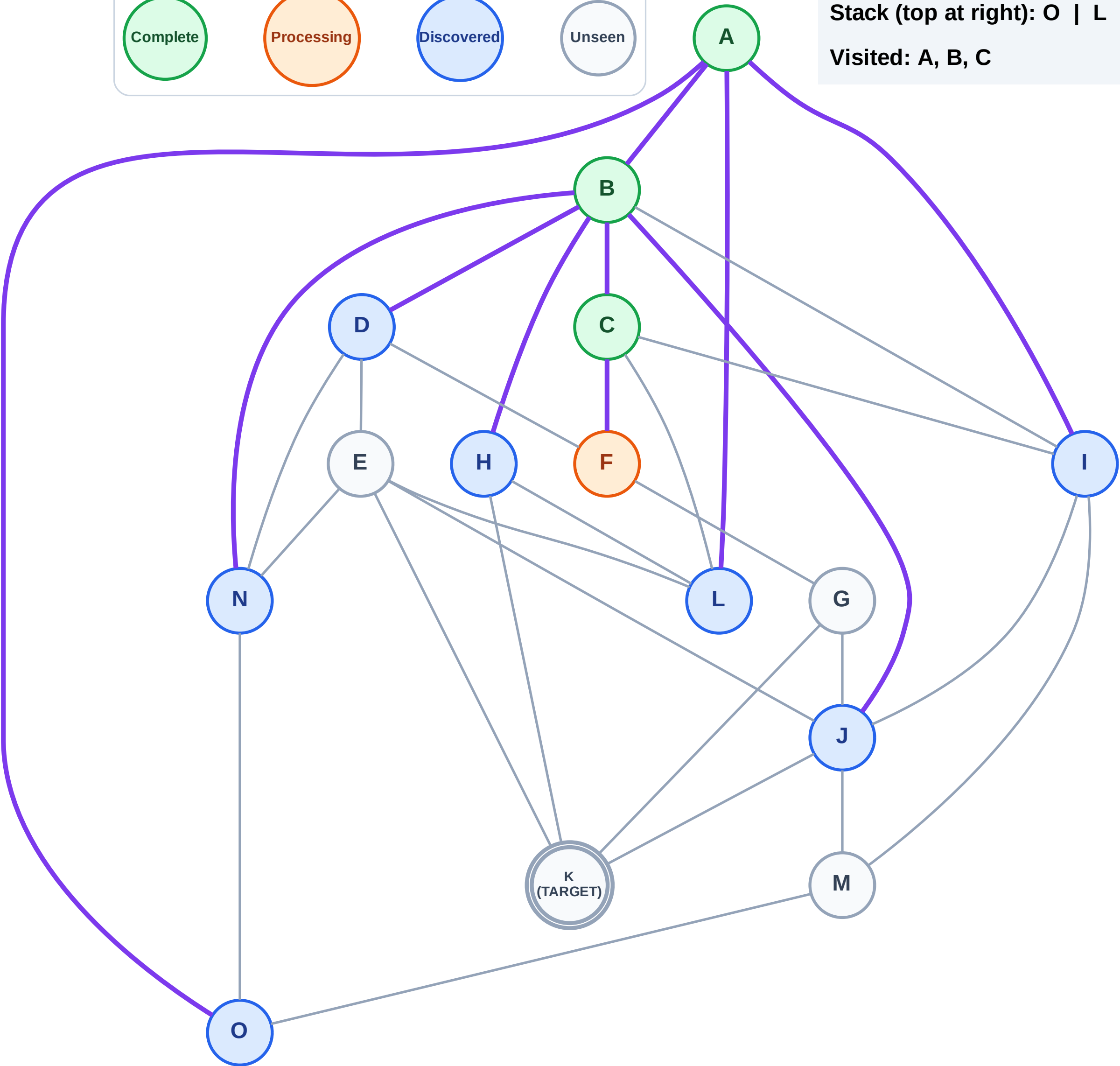
Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

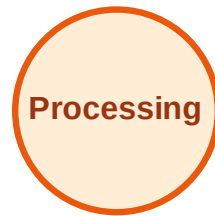
Visited: A, B, C



DFS Traversal

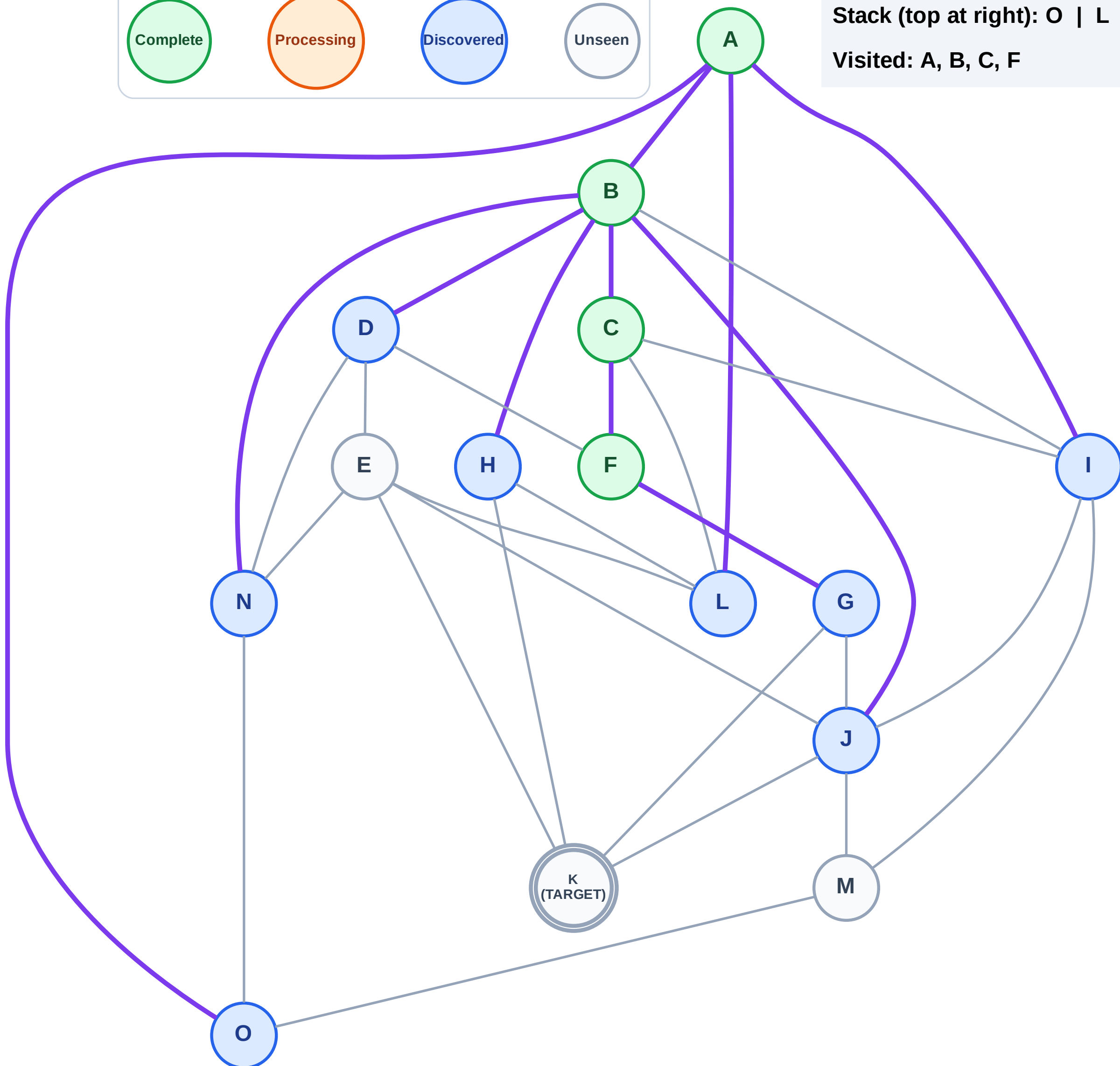
Step 9: Discover G from F

Legend



Stack (top at right): O | L | I | N | J | H | D | G

Visited: A, B, C, F



DFS Traversal

Step 10: Process node G

Legend

Complete

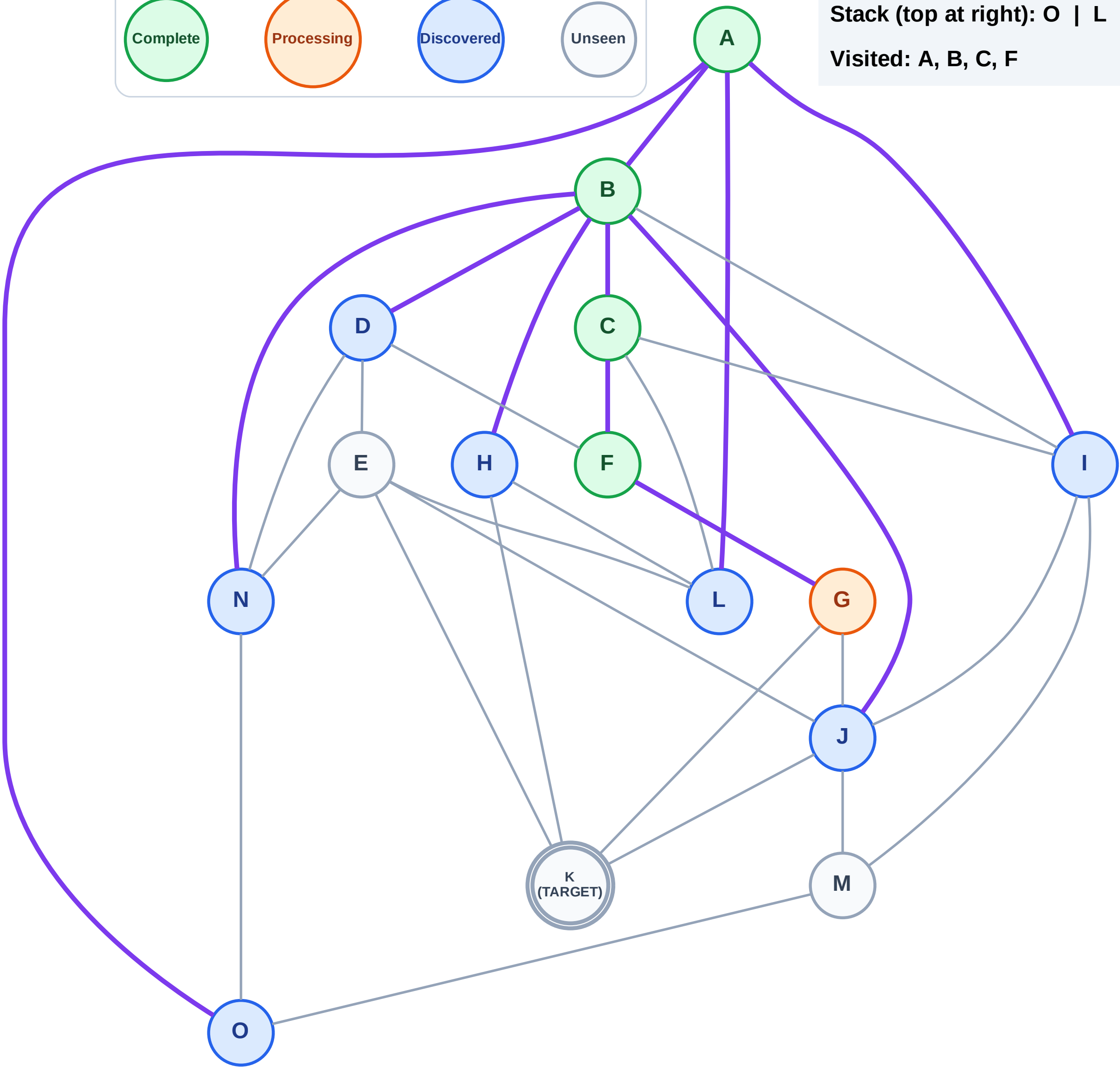
Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

Visited: A, B, C, F



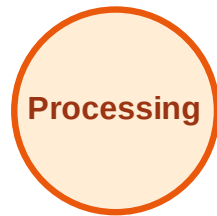
DFS Traversal

Step 11: Discover K from G

Legend



Complete



Processing



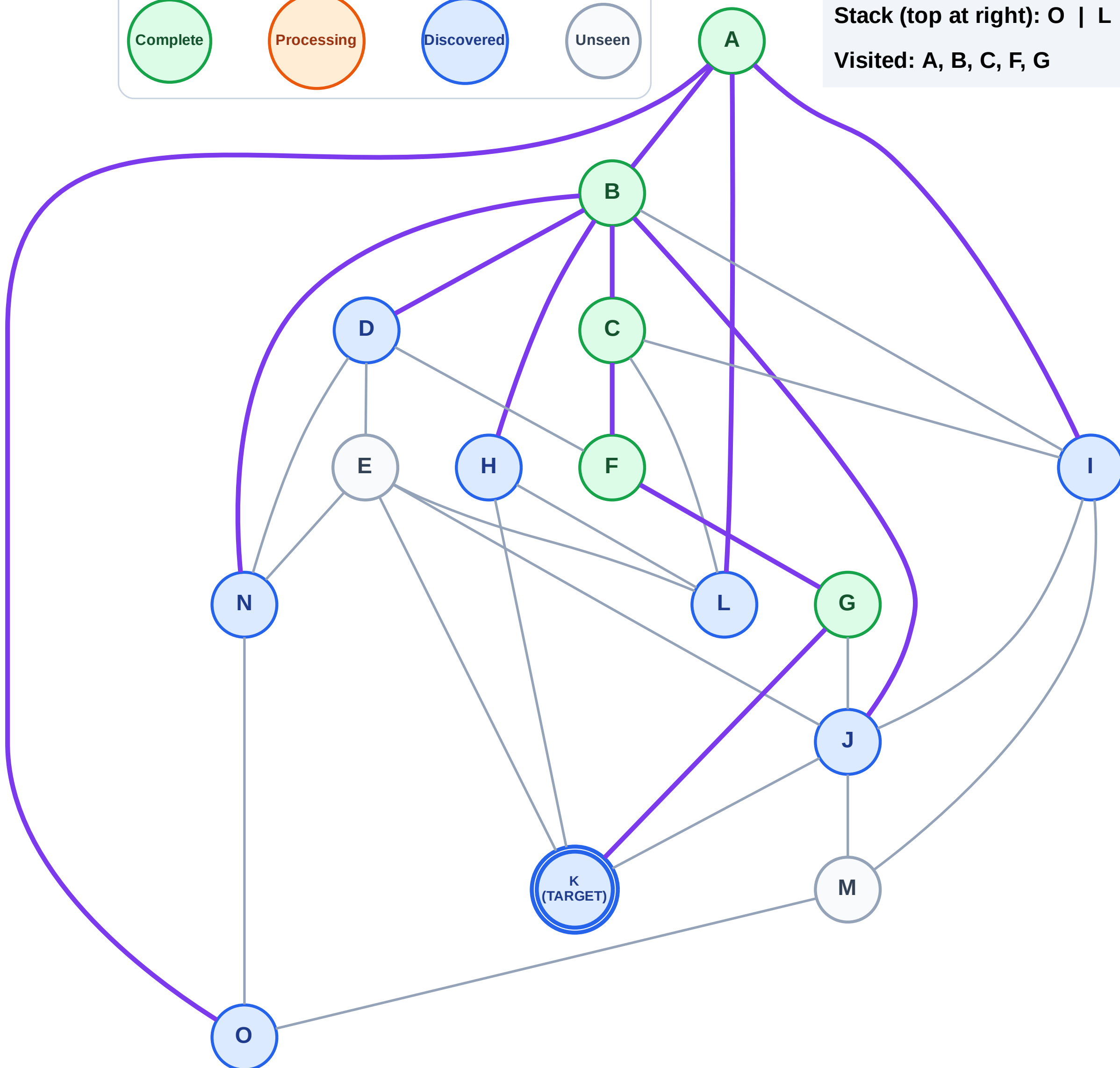
Discovered



Unseen

Stack (top at right): O | L | I | N | J | H | D | K

Visited: A, B, C, F, G



DFS Traversal

Step 12: Process node K

Legend

Complete

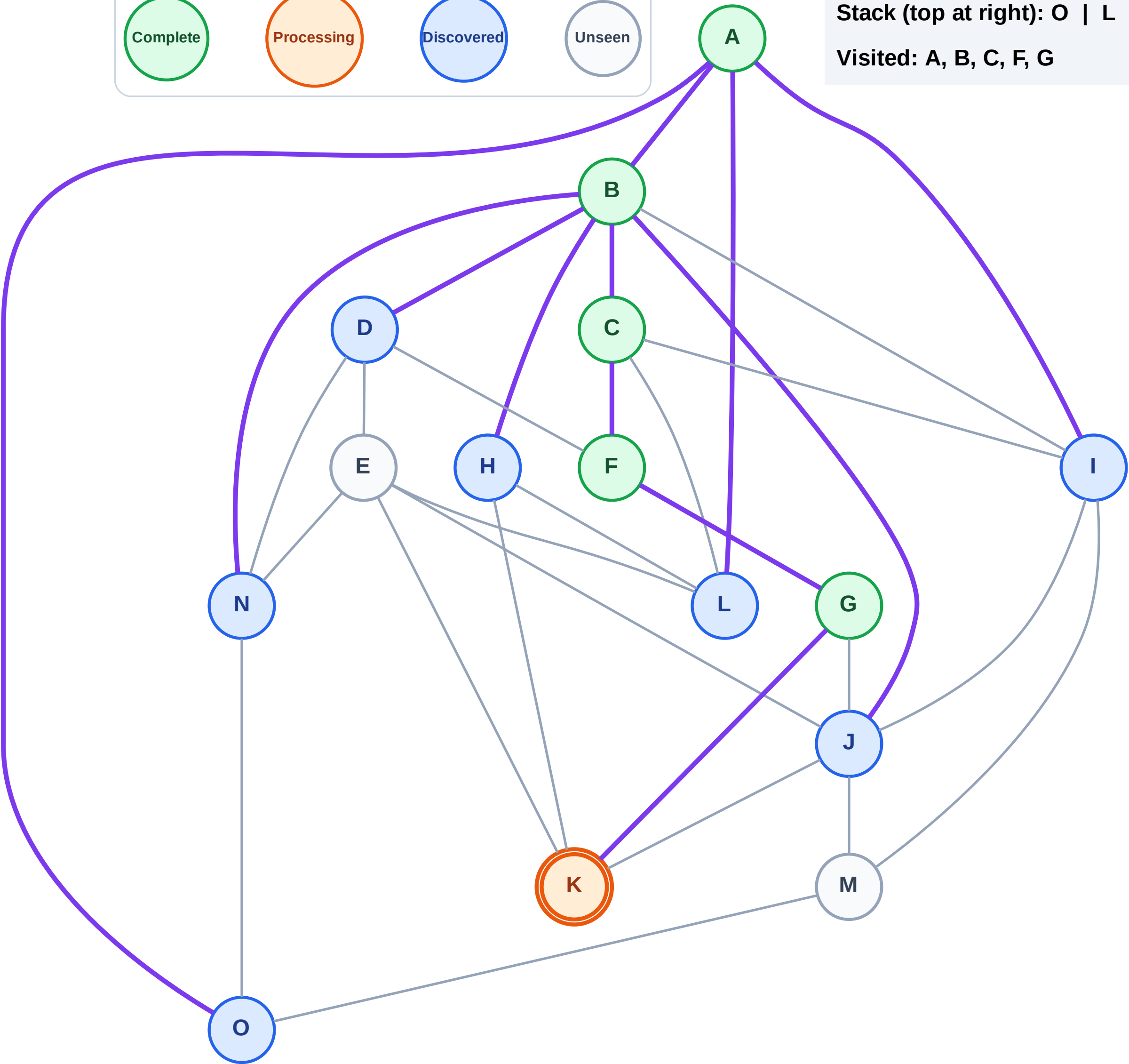
Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

Visited: A, B, C, F, G



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (5 hops): A → B → C → F → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | L | I | N | J | H | D

Visited: A, B, C, F, G, K

